



DIGITAL LEARNING CONTENT MANUAL

GRADE TWELVE

ENGLISH

MATHS

BIOLOGY

CHEMISTRY

PHYSICS

URDU

PAKISTAN STUDIES

Table of Contents

1. Introduction	2
2. Available digital content from Grade KG - Grade 12	3
3. How to read video codes	4
4. Grade 12 Digital Content Alignment with SLOs	5
a. English	5
b. Maths.....	10
c. Biology.....	13
d. Chemistry.....	29
e. Physics	38
f. Urdu.....	48
g. Pakistan Studies	57
5. How to Use This Document in Class	60
6. Ethical and Copyright Considerations.....	61
7. Conclusion.....	62

Digital Content Practical Lookup Guide

1. Introduction

The Ministry of Federal Education and Professional Training is dedicated to enhancing the quality of distance learning and creating a comprehensive educational platform that serves a diverse group of learners across Pakistan. As part of this effort, the Ministry has procured digital content, including audio and video resources, that align with the Pakistan National Curriculum and Student Learning Outcomes (SLOs).

This guide is designed to help teachers navigate this digital content efficiently. Every video has been assigned a unique code, and this document organizes them in a way that clearly shows which video supports which SLO, for which grade and subject. Each table includes the video code, topic, SLO(s) covered, textbook unit, and duration.

Teachers can use this guide for lesson planning, in-class instruction, or revision sessions. It ensures that digital resources are used effectively, keeping lessons aligned with the curriculum and maximizing learning outcomes. The structure is consistent across all grades and subjects, making it easy to locate the right video at the right time.

This guide is intended as a practical companion to support teachers in delivering high-quality, curriculum-aligned instruction through digital content.

2. Available digital content from Grade KG- Grade 12

Digital Learning Content (Video)

Grade	Subject	No. of Packages	Total No. of Packages
KG	Math	21	133
	English	36	
	Urdu	25	
	General Knowledge	51	
Grade 1	Math	35	147
	English	33	
	Urdu	60	
	General Knowledge	19	
Grade 2	Math	45	166
	English	71	
	Urdu	16	
	General Knowledge	34	
Grade 3	English	92	176
	General Knowledge	29	
	Maths	39	
	Urdu	16	
Grade 4	English	92	190
	Math	36	
	Science	42	
	Urdu	20	
Grade 5	English	73	155
	Math	23	
	Science	34	
	Urdu	25	

Grade	Subject	No. of Packages	Total No. of Packages
Grade 6	Maths	31	120
	Science	27	
	English	37	
	Computer Science	25	
Grade 7	English	64	167
	Maths	51	
	Science	24	
	Computer Science	28	
Grade 8	English	60	162
	Maths	45	
	Science	25	
	Computer Science	32	

Grade	Subject	No. of Packages	Total No. of Packages
Grade 9	Biology	19	119
	Chemistry	21	
	English	20	
	Maths	42	
Grade 10	Physics	17	131
	Biology	26	
	Chemistry	12	
	English	21	
Grade 11	Maths	55	552
	Physics	17	
	Biology	110	
	Chemistry	94	
Grade 12	English	55	544
	Maths	88	
	Physics	91	
	Urdu	114	
ECD – Grade 12	Biology	103	40
	Chemistry	86	
	English	55	
	Maths	67	
ECD – Grade 12	Pak. Studies	45	40
	Physics	56	
	Urdu	132	
	Gamified Educational Content	40	

Total no. of Video Packages	2802
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Sesame, Idara Taleem o Agahi (ITA), Environment & Entrepreneurship

Sesame Workshop	
Social Emotional Learning (SEL)	17
Maths	38
Science/ General Knowledge	21
Total	76 videos 55 (5 min videos) and 21 (27 minute videos)

ITA - Storytelling	
Grade 1	6
Grade 2	6
Grade 3	7
Total	19

Theme	No. of Packages
Environments for Living Lot 1 (ECD–2)	6
Environments for Living Lot 2 (Grades 3–5)	18
Environments for Living Lot 3 (Grades 6–8)	21
Entrepreneurship (Grades 6–8)	20

3. How to read video codes

Every video in this digital content collection has a unique code. The code helps teachers quickly identify:

- the grade
- the subject
- the textbook unit
- the package and part of the video

Understanding the code allows you to find the right video quickly without searching through multiple files.

Video Code Format

Example: **G01EnU01P01a**

- G01 = Grade 1
- En = English
- U01 = Unit 1
- P01a = Package 1, Part a

Code Reference Table:

Grade	Subject	Unit	Package
G01-G12, EY	M (Mathematics)	U01	P01
G01-G12, EY	Ur (Urdu)	U01	P01
G01-G12, EY	En (English)	U01	P01
G01-G12, EY	Sc (Science)	U01	P01
G01-G12, EY	Ph (Physics)	U01	P01
G01-G12, EY	Bi (Biology)	U01	P01
G01-G12, EY	Ch (Chemistry)	U01	P01
G12	PS (Pakistan Studies)	U01	P01

Notes for Teachers:

- Use this table to decode video codes quickly.
- “EY” refers to Early Years/Kindergarten.
- Package numbers indicate video sequence within a unit.
- Always cross-check the mapping tables in this guide for SLO coverage before showing a video in class.

4. Grade 12 Digital Content Alignment with SLOs

a. English

Unit	SLO	Package Code	Package Name
Competency 1: Reading and Thinking skills	➤ Apply strategies to comprehend questions by marking key words, verbs and tenses in a variety of question types: i. Literal/ textual/ factual ii. Interpretive iii. Inferential iv. Evaluative v. Personal response vi. Open ended ➤ Respond orally and in writing.	G12EnU01P01	Comprehension
	➤ Use dictionaries to • locate guide words. • locate entry word. • choose appropriate word definition. • identify pronunciation with the pronunciation key. • identify syllable division, and stress pattern. • identify parts of speech. • identify correct spellings. • identify phrases through key words. • recognize abbreviations used in a dictionary. • locate phrases and idioms. • comprehend notes on usage. • identify word etymology. ➤ Locate appropriate synonyms and antonyms in a thesaurus. ➤ Utilize appropriate informational sources including encyclopedias and internet sources. ➤ Use library skills to • alphabetize book titles, words and names. • locate fiction and nonfiction books / books by subject. • understand card catalogue. • locate and using card catalogue. • identify three kinds of catalogue cards i.e. author card, title card, subject card. • use Dewey decimal system. • use numbers on books and catalogue cards. • use case and shelf labels in the library. • use the reference section in the library. • use computer catalogue. ➤ Utilize effective study strategies e.g. note taking / note making, writing a summary, creating a mind map to organize ideas. ➤ Use textual aids such as table of contents, footnote, glossary, preface, sub headings etc. to	G12EnU01P02	Correction of Sentences Part 1
		G12EnU01P03	Correction of Sentences Part 2
		G12EnU01P04	Correction of Sentences Part 3
		G12EnU01P05	Correction of Sentences Part 4
		G12EnU01P06	Correction of Sentences Part 5
		G12EnU01P07	Correction of Sentences Part 6
		G12EnU01P08	Correction of Sentences Part 7
		G12EnU01P09	Correction of Sentences Part 8
		G12EnU01P10	Correct Spellings

	- comprehend texts. - identify and select relevant information in a book.		
Competency 2: Writing Skills	- Develop focus for his or her writing. - Select and use a variety of pre-writing strategies such as brainstorming, mind mapping, outlining etc. - Plan draft and revise writing to ensure that it - is focused, purposeful, includes a sense of audience, and shows insight into the writing situation. - has an appropriate writing style (expository, narrative, etc.) for a given purpose. - has an organizational pattern that reflects a clear overall progression of ideas through proper use of signal and reference words. - uses writing strategies as are appropriate to the purpose of writing. - has varied sentence structure and length. - has a good command of language with precision of expression. - Proof read and edit their own, peers', and given texts for errors of usage and style: - Faulty sentence structure. - Unclear pronoun reference. - Incomplete comparison. - Misplaced modifiers. - Dangling modifiers. - Subject / verb agreement. - Inconsistencies in verb / tense. - Faulty parallelism. - Confusion of adjectives and adverbs. - Wordy phrases. - Redundancy. - Vague language. - Inappropriate diction. - Clichés. - Conventions of format. - Errors of punctuation and spelling	G12EnU02P11	Subject Verb and Noun Agreement
		G12EnU02P12	Noun Pronoun Agreement and Modifiers
		G12EnU02P13	Parallelism and Wordiness
		G12EnU02P14	Essays Part 1
		G12EnU02P15	Essays Part 2
		G12EnU02P16	Essays Part 3
		G12EnU02P17	Essays Part 4
		G12EnU02P18	Essays Part 5
Competency 4: Formal and Lexical Aspects of Language.	- Use the pronunciation key to pronounce words with developing accuracy. - Recognize silent letters in words and pronounce them with developing accuracy. - Recognize, pronounce and represent primary and secondary stress in words with the help of a dictionary. - Use appropriate vocabulary and correct spelling in their own writing: - Illustrate the use of dictionary for finding appropriate meaning and correct spellings. - Use a thesaurus to locate the synonyms closest to the meaning of the given word in the context.	G12EnU04P19	Pronunciation
		G12EnU04P20	Idioms and Phrasal Verbs
		G12EnU04P21	Idioms Part 1
		G12EnU04P22	Idioms Part 2

• Examine and interpret transitional devices that show comparison, contrast, reason, concession, condition, emphasis. • Deduce the meaning of unfamiliar words from the context using contextual clues. • Use the knowledge of roots, suffixes and affixes to determine the meaning of unfamiliar words. • Understand and use colloquial and idiomatic expressions given in the text / glossary. • Explore the use of synonyms with varying shades of meaning used for various purposes e.g. propaganda, irony, parody and satire. • Use various reference sources to refine vocabulary for interpersonal, academic and work place situations, including figurative, idiomatic and technical vocabulary.	G12EnU04P23	Idioms Part 3
	G12EnU04P24	Idioms Part 4
	G12EnU04P25	Idioms Part 5
	G12EnU04P26	Idioms Part 6
	G12EnU04P27	Idioms Part 7
	G12EnU04P28	Idioms Part 8 and Vocabulary
	G12EnU04P29	Vocabulary, Morphology and Greek Root Knowledge
	G12EnU04P30	Punctuation Part 1
	G12EnU04P31	Punctuation Part 2
	G12EnU04P32	Punctuation Part 3

	Brackets) to enclose explanation, comment or criticism inserted by someone other than the person quoted. ➤ Recognize and use omission marks or ellipses to signify the omission or deletion of letters or words in sentences.		
	➤ Demonstrate use of collective, countable and uncountable, material and abstract nouns.	G12EnU04P33	Nouns and It's Types Part 1
	➤ Demonstrate use of nouns, noun phrases and clauses in apposition.	G12EnU04P34	Nouns and It's Types Part 2
	➤ Apply rules of change of number of nouns.	G12EnU04P35	Nouns and It's Types Part 3
	➤ Recognize and demonstrate use of words that have double plurals	G12EnU04P36	Phrases and Clauses
	➤ Illustrate use of pronouns.	G12EnU04P37	Pronouns and It's Types Part 1
	➤ Identify, and demonstrate use of relative pronouns.	G12EnU04P38	Pronouns and It's Types Part 2
	➤ Recognize rules for using indefinite pronouns.		
	➤ Illustrate use of pronoun-antecedent agreement.	G12EnU04P39	Pronouns and It's Types Part 3
	➤ Illustrate use of cataphoric and anaphoric references.		
	➤ Apply rules for the use of a, an and the, wherever applicable in speech and writing	G12EnU04P40	Articles Part 1
		G12EnU04P41	Articles Part 2 and Verbs
	Illustrate the use and all functions of modal verbs. ➤ Illustrate use of regular and irregular verbs. ➤ Illustrate use of transitive and intransitive verbs. ➤ Make and use present and past participles. ➤ Identify, recognize the function and use of perfect participles. ➤ Illustrate the use of infinitives and infinitive phrases. ➤ Illustrate the use of gerunds and gerund phrases	G12EnU04P42	Types of Verbs
	➤ Illustrate use of tenses.	G12EnU04P43	Tenses Part 1
	➤ Identify, change the form of, and use Future Continuous Tense.		
	➤ Identify form and use Future Continuous, Future Perfect and Future Perfect Continuous Tenses.	G12EnU04P44	Tenses Part 2
	➤ Classify adjectives into different types. Change and use degrees of adjectives.	G12EnU04P45	Adjectives Part 1
	➤ Follow order of adjectives in sentences.	G12EnU04P46	Adjectives Part 2
	➤ Identify and use adjective phrases and clauses.	G12EnU04P47	Adjectives Part 3
		G12EnU04P48	Adjective Clause and Adjective Phrase
	➤ Illustrate use of adverbs.		
	➤ Recognize varying positions of adverbs in sentences according to their kinds and importance.	G12EnU04P49	Adverbs

	➤ Identify and use adverbial phrases and clauses		
	➤ Illustrate use of prepositions of position, time and movement and direction	G12EnU04P50	Prepositions Part 1
		G12EnU04P51	Prepositions Part 2
	➤ Use in speech and writing, all the appropriate transitional devices	G12EnU04P52	Transitional Devices
	➤ Analyze sentences for different clauses and phrases; evaluate how their positions in sentences change meaning and affect communicative function.	G12EnU04P53	Sentence Structure
	➤ Recognize and use sentence inversion for various purposes. ➤ Analyze and construct simple, compound and complex sentences. ➤ Identify, analyze and construct conditional sentences.	G12EnU04P54	Sentences and It's Types
		G12EnU04P55	Conditionals and Inversions

b. Maths

Unit	SLO	Package Code	Package Name
Functions and limits	Revision of the work done in the previous classes	G12MU01P01	Function and Its Value
	Kinds of functions	G12MU01P02	Vertical Line Test and Types of Function
		G12MU01P03	Types of Functions and Examples
		G12MU01P04	Domain and Range
		G12MU01P05	Exercise Questions
		G12MU01P06	Composition of Functions
	Composition and inverse of function	G12MU01P07	Inversion of Functions
		G12MU01P08	Limits
	Limits of important functions	G12MU01P09	Limits of Important Functions
	Continuous and Discontinuous Functions	G12MU01P10	Continuous and Discontinuous Functions
Differentiation	Introduction	G12MU02P11	Concept of Differentiation
	Differentiation of Algebraic Expression		
	Theorems of Differentiation	G12MU02P12	Product and Quotient Rule
	Differentiation of Functions other than algebraic + Successive differentiation	G12MU02P13	Derivative of $\sin x$ and $\tan x$
		G12MU02P14	Derivative of $\cos x$ and Examples
		G12MU02P15	Derivative of Inverse Trigonometric Functions
		G12MU02P16	Derivative of Exponential Log Function
		G12MU02P17	Derivative of Log Function Part 2 and Examples
		G12MU02P18	Hyperbolic, $\tanh x$ and Inverse Hyperbolic Function
	Maclaurin's and Taylor's Theorems	G12MU02P19	Maclaurin Series
	Extreme values	G12MU02P20	Increasing and Decreasing Function
		G12MU02P21	Relative Extrema
Integration	Introduction to Integration	G12MU03P22	

	Theorems on Anti-derivatives		Introduction to Integration
	Definite integrals	G12MU03P23	Integrals
	Application of definite integral	G12MU03P24	Application of Definite Integral
	Differential Equation	G12MU03P25	Differential Equations
Introduction to Analytic Geometry	Coordinate system	G12MU04P26	Distance and Ratio Formula
	Translation and Rotation of axes	G12MU04P27	Translation and Rotation of Axes
	Equations of straight lines	G12MU04P28	Slope of Line
		G12MU04P29	Equation of Straight Line
		G12MU04P30	Exercise and Examples Part 1
		G12MU04P31	Exercise and Examples Part 2
		G12MU04P32	Exercise and Examples Part 3
	Two and three straight lines	G12MU04P33	Concurrency, Angle and Area
	Homogenous equations of 2nd degree in two variables x and y	G12MU04P34	Second Degree Homogenous Equation
Linear Inequalities and Linear Programming	Linear Inequalities & their graphs	G12MU05P35	Introduction to Linear Inequalities
		G12MU05P36	Graphing of Linear Inequality
	Feasible solution set	G12MU05P37	Corner Points of Feasible Region
	Linear Programming	G12MU05P38	Linear Programming
Conic sections	Introduction	G12MU06P39	Conic Section and Equation of Circle
	Circle		
	Tangents and normals	G12MU06P40	Tangent and Normals
		G12MU06P41	Tangent and Normals Questions Part 1
		G12MU06P42	Tangent and Normals Questions Part 2
	Analytic proofs of important properties of circle	G12MU06P43	Proofs of Properties of Circle
	Parabola and its elements	G12MU06P44	Parabola and Its Equation
		G12MU06P45	Elements of Parabola
		G12MU06P46	Elements of Parabola Part 2
	Equation of parabola with given elements	G12MU06P47	Equation of Parabola with Given Elements

	Tangents and normals to parabola	G12MU06P48	Tangents and Normals to a Parabola
	Ellipse and its elements	G12MU06P49	Ellipse and Its Elements
	Equation of an ellipse with its elements	G12MU06P50	Equation of Ellipse with Its Elements
	Tangents and normal to ellipse	G12MU06P51	Tangents and Normals to Ellipse
	Equation of hyperbola with its elements	G12MU06P52	Equation of Hyperbola, Tangents and Normals to Hyperbola
	Tangents and normals to hyperbola		
Vectors	Introduction to vector in a plane	G12MU07P53	Vectors and their Geometric Interpretation
		G12MU07P54	Addition of Vectors
		G12MU07P55	Questions Related to Vectors in a Plane
		G12MU07P56	Ratio Formula
	Introduction to vector in space	G12MU07P57	Vector in Space
	Scalar Product of two vectors	G12MU07P58	Scalar Product
		G12MU07P59	Dot product and Projection of One Vector upon Other
		G12MU07P60	Questions Related to Scalar Product of Two Vectors Part 1
		G12MU07P61	Questions Related to Scalar Product of Two Vectors Part 2
	Vector Product of two vectors	G12MU07P62	Vector Product
		G12MU07P63	Analytic Expression of Cross Product
		G12MU07P64	Questions Related to Vector Product of Two Vectors Part 1
		G12MU07P65	Questions Related to Vector Product of Two Vectors Part 2
	Scalar Triple Product of Vectors	G12MU07P66	Scalar Triple Product, Parallelepiped and Tetrahedron

		G12MU07P67	Questions Related to Scalar Triple Product of Vectors
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c. Biology

Unit	SLO	Package Code	Package Titles
HOMEOSTASIS	Define osmoregulation.	G12BiU01P01	Homeostasis and Concept of Feedback
	Describe three elements i.e. receptors, control center and effectors which operate homeostatic mechanisms.		
	Relate the homeostatic mechanisms with the negative and positive feedback systems.		
	Differentiate between osmoconformers and osmoregulators.	G12BiU01P02	Osmoregulatory Adaptation and Problems Part 1
	Explain the problems faced by osmoregulators.		
	Explain the different methods of osmoregulation found in freshwater, marine water and terrestrial habitats.	G12BiU01P03	Osmoregulatory Adaptation and Problems Part 2
	List various nitrogenous compounds excreted during the process of excretion.	G12BiU01P04	Excretory Products and Their Correlation With Habitat
	Explain the nature of excretory products in relation to habitat.		
	Explain different organs of urinary system. Describe the structure of kidney and relate it with its function.	G12BiU01P05	Excretory System of Man
	Explain the detailed structure of nephron.		
	Justify the functioning of kidneys as both excretion and osmoregulation.	G12BiU01P06	Functioning of Kidneys
	Explain the processes of glomerular filtration, selective re-absorption and tubular secretion as the events in kidney functioning.	G12BiU01P07A	Regulatory Functions of Kidneys
	Compare the function of two major capillary beds in kidneys i.e. glomerular capillaries and peritubular capillaries.		

	Explain that concentration of urine is regulated by counter-current and hormonal mechanisms.		
	List urinary tract infections and the bacteria responsible.	G12BiU01P07B	Disorders of the Urinary Tract Part 1
	Explain the causes and treatments of kidney stones.		
	Outline the causes of kidney failure.	G12BiU01P08	Disorders of the Urinary Tract Part 2
	Explain in detail the mechanism and problems related to dialysis.		
	Describe the principles and the problems associated with kidney transplant.		
	Define thermoregulation and explain its needs.	G12BiU01P09	Thermoregulation
	Classify animals on the basis of the source of body's heat i.e. ectotherms and endotherms.		
	Classify the animals on the bases of the ability to thermoregulate i.e. poikilotherms and homeotherms.		
	Describe the regulatory strategies in man for thermoregulation.		
SUPPORT AND MOVEMENT	Describe the structure of bone and compare it with that of cartilage.	G12BiU02P01	Bone and Cartilage
	Explain the functions of osteoblasts, osteoclasts and osteocytes.		
	Identify the main divisions of human skeleton.	G12BiU02P02	Divisions of the Human Skeleton and Axial Skeleton
	List the bones of appendicular and axial skeleton of man.	G12BiU02P03	Appendicular and Axial Skeleton
	Describe three types of joints i.e. fibrous joints, cartilaginous joints and synovial joints and give example of each.	G12BiU02P04	Joints
	Describe the disorders of human skeleton (disc-slip, spondylosis, sciatica, arthritis) and their causes.	G12BiU02P05	Disorders of Human Skeleton
	State different types of fractures (simple, compound, and complicated) and describe the the repair process of simple fractures.	G12BiU02P06	Disorders of Skeleton
	Describe the injuries in joints (dislocation and sprain) and their first aid treatment.		

	Describe the first-aid treatment for fracture.		
	Compare smooth muscles, cardiac muscles and skeletal muscles.	G12BiU02P07	Muscles and Its Types
	Explain the ultra-structure of the skeletal muscle.	G12BiU02P08	Ultra-Structure of the Skeletal Muscle
	Explain the sliding filaments model of muscle contraction.	G12BiU02P09	Muscle Contraction Sliding Filament Model
	Describe the action of antagonistic muscles in the movement of knee joint.	G12BiU02P10	Antagonistic Muscles and Muscle Disorders
	Explain muscle fatigue, cramps and tetany.		
	Differentiate between tetanus and muscle tetany.		
NERVOUS COORDINATION	Explain the structure and functioning of the receptors for smell, taste and touch/ pain.	G12BiU03P01	Receptors and Neurotransmitters
	Recognize receptors as transducers sensitive to various stimuli.		
	Classify neurotransmitters as inhibitory and excitatory and list some common examples.		
	Trace the path of a message transmitted to the CNS for processing.	G12BiU03P02	Nervous Coordination
	Identify the three neurons (sensory, intermediate, motor) involved in nervous transmission.		
	Describe the detailed structure of a sensory neuron, associative and a motor neuron and relate the specialization in structures with functions.		
	Differentiate between myelinated and non-myelinated neurons.	G12BiU03P05	Neurons
	Explain the function of the three types of neurons with the help of a reflex arc.	G12BiU03P06	Reflex Action
	Define nerve impulse.	G12BiU03P07A	Nerve Impulse Part 1
	Explain synaptic transmission of nerve impulse.		
	Describe the generation and transmission of nerve impulse.		
	Name the factors responsible for the resting membrane potential of neuron.		
	Evaluate from a graph the phenomena of polarization, depolarization and hyperpolarisation of membrane.		

	Describe the structure of synapse.	G12BiU03P07B	Nerve Impulse Part 2
	Identify the main components of the nervous system.	G12BiU03P08	Human Nervous System Part 1
	Explain briefly the functions of major divisions of brain.		
	Describe the architecture of human brain and compare its sectional view with that of the spinal cord.		
	Describe cranial and spinal nerves in man.	G12BiU03P09	Human Nervous System Part 2
	Explain the structure, types and functions of autonomic nervous system.		
	Define narcotic drugs as agents that interact with the normal nervous activity.	G12BiU03P10	Effects of Drugs on Nervous Coordination
	Compare the use and abuse of drugs with respect to heroine, Cannabis, nicotine, alcohol and inhalants like nail polish remover and glue.		
	Associate the effects of drug addiction and tolerance with the functioning of nervous system.		
	Define and explain the terms; drug addiction and drug tolerance with reference to caffeine and nicotine.		
	Describe withdrawal symptoms of alcohol.		
	Classify nervous disorders into vascular, infectious, structural, functional, and degenerative disorders.	G12BiU03P11	Vascular and Infectious Nervous Disorders
	Describe the causes, symptoms and treatment one type of each category of disorders outlined above. (e.g., stroke as vascular, meningitis as infectious, brain tumor as structural, headache as functional, and Alzheimer disease as degenerative disorder).	G12BiU03P12	Structural, Functional, and Degenerative Nervous Disorders
	Explain the principles of the important diagnostic tests for nervous disorders i.e. EEG, CT scan and MRI.	G12BiU03P13	Diagnostic Tests for Nervous Disorders
CHEMICAL COORDINATION	State the role of hormones as chemical messengers.	G12BiU04P01	Hormones the Chemical Messengers
	Describe the chemical nature of hormones and correlate it with important hormones.		

	Trace the path of the chemical message from its release from the endocrine gland to its action at the target site.		
	Explain the two modes of hormone action at the cells of target site.		
	Locate the following endocrine glands in human body; pituitary, thyroid, parathyroid, pancreas, adrenal, gonads.	G12BiU04P02	Endocrine System of Man and Hypothalamus
	Explain the neurosecretory role of hypothalamus.	G12BiU04P03	Endocrine System of Man and Pituitary Gland
	Name the hormonal secretions of the above-mentioned glands.	G12BiU04P04	Endocrine System of Man, Thyroid, and Parathyroid Gland
	Outline the major functions of the hormones of above mentioned glands and also relate the problems associated with the imbalance of these hormones.	G12BiU04P05	Endocrine System of Man and Pancreas
	Describe the functions of the hormones secreted by the endocrine tissue other than the mentioned above.	G12BiU04P06	Endocrine System of Man and Adrenal Gland
		G12BiU04P07	Gonads and Other Endocrine Tissues, Cells
	Describe positive feedback with reference to Oxytocin and negative feedback with reference to Insulin and Glucagon.	G12BiU04P08	Feedback Mechanism
	Outline the concept of Feedback mechanism of hormones.		
BEHAVIOR	Define behavior as the series of activities performed by an organism in response to stimuli.	G12BiU05P01	Nature of Behavior
	Explain relationship between stimuli and behavior.		
	Describe the relationship between heredity and behavior.		
	Explain, through examples, the biological rhythms.		
	Define innate (inborn) behavior.	G12BiU05P02	Innate Behavior and Social Behavior
	Define instincts and justify these as a type of innate behavior.		

	Justify the fact that each species displays its own characteristic instinctive behavior through following examples; migration of salmon, dances of bees, construction of hanging nest by birds, construction of intricate web by spider and mating behavior of stickleback fish.		
	Define learning and distinguish between learning and innate behavior.		
	Justify reflexes as a type of innate behavior, by giving examples from man and invertebrates.		
	Describe examples of innate behavior in terms of taxis shown by unicellular organisms and tropism shown by plants.		
	Differentiate between animal aggregations and animal societies.		
	Describe social behavior in terms of hostile and helpful interactions between animals belonging to the same species.		
	Describe agonistic behavior and relate it with the maintenance of social order in terms of territories and dominance hierarchies.		
	Explain territorial behavior by quoting example of the territories of monkeys and gorillas.		
	Explain dominance hierarchy by quoting example of the pecking order of chicken.		
	Define altruism and illustrate it through the organization of a honeybee society.		
	Define habituation and illustrate it through the example of squirrels' adjustment in a park.	G12BiU05P03	Learned Behavior
	Explain imprinting by narrating the work of Lorenz.		
	Differentiate habituation and imprinting as reversible and irreversible learned behaviors.		
	Describe classical conditioning by narrating the work of Pavlov on salivary reflex in dogs.		
	Describe instrumental conditioning (trial-and-error learning) by narrating the work of Skinner on rats' learning.		

	Describe latent learning, through the example of a rat in a maze with no reward.		
	Interpret Kohler's work on chimpanzee's insight learning to justify that reasoning and planning are involved in the insight learning.		
REPRODUCTION	Describe the structures of male reproductive system identifying their functions.	G12BiU06P01	Male Reproductive System Part 1
	Explain the principal reproductive hormones of human male and explain their role in the maintenance and functioning of reproductive system.	G12BiU06P02	Male Reproductive System Part 2
	Explain the structures of female reproductive system and describe their functions.	G12BiU06P03	Female Reproductive System
	Describe the menstrual cycle emphasizing the role of hormones.	G12BiU06P04	Menstrual Cycle
	Describe the causes of female and male infertility.	G12BiU06P05	Disorders of Reproductive System
	Explain that in-vitro fertilization (test tube babies) is one of the methods to solve the problem of infertility.	G12BiU06P06	Treatment of Infertility
	Define miscarriage and state its causes.	G12BiU06P07	Miscarriage
	Relate miscarriage with abortion.		
	Describe the causes, symptoms and treatment of gonorrhea and syphilis.	G12BiU06P08	Sexually Transmitted Diseases
	Explain AIDS as a worldwide sexually transmitted disease.		
DEVELOPMENT AND AGING	Describe cleavage and relate it with amount of yolk.	G12BiU07P01	Embryonic Development Part 1
	Define organogenesis.		
	Explain the events of gastrulation.	G12BiU07P02	Embryonic Development Part 2
	List the tissues and organs formed from the three germ layers.		
	State the events of neurulation.		
	Describe the formation of neural crest and list the structures that are derived from neural crest cells.	G12BiU07P03	Embryonic Development Part 3
	Through experimental narration, describe the role of the nucleus	G12BiU07P04	Control of Development

	and cytoplasm in controlling development.		
	Differentiate the terms gestation and pregnancy.	G12BiU07P06	Pregnancy, Twins and Quadruplets, Birth and Nursing
	Describe the events of development in human in terms of first, second and third trimesters.		
	Describe in brief the development of twins and quadruplets.		
	Define colostrum and describe the role of prolactin in the production and of oxytocin in the secretion of milk.		
	State the hormonal regulation in the end of milk production.		
	Describe the role of fetal and maternal hormones in initiating labor pains.		
	Define the term premature birth and correlate it with the growth phases in the second and third trimesters.	G12BiU07P07	Placenta, Birth, Nursing, and Disorders
	Describe the structural details of placenta and umbilical cord.		
	Define afterbirth and describe how umbilical cord is detached from the baby. culminating in the birth of baby		
	Define Colostrum. Compare breast-feeding and bottle-feeding, in terms of advantages and disadvantages.		
	Describe the maternal derived abnormalities (rubella, abnormal neural tube, thyroid gland and limb development).		
	Relate the major genetic abnormalities in embryos with spontaneous abortion.		
	Describe how fetal surgery helps to correct the detected fetal developmental problems.		
	Define the term allometric growth and correlate it with the postnatal development in human.	G12BiU07P08	Postnatal Development in Human
	Define the term aging.	G12BiU07P09	Aging
	Rationalize aging as a part of normal development.		
	List the genetic and extrinsic factors responsible for aging.		

	State the changes (graying, thinning hair, pigmented patches of skin, slowed movements, fading vision, impaired hearing, reduced ability to adapt to stress and decreased resistance to infections) as primary aging.		
	State the changes that are the result of environmental, lifestyle factors such as disease, disuse (lack of exercise), and abuse (smoking, obesity, malnutrition, and exposure to ultra-violet light) as secondary aging.		
	List some changes that occur at the system and those that occur at cellular level during aging.		
INHERITANCE	Associate inheritance with the laws of Mendel.	G12BiU08P01	Mendelian Inheritance
	Express limitations in the law and its usefulness.		
	State the scope of independent assortment in variation.		
	Evaluate that inheritance of genes and their mixing during fertilization is based on mathematical probabilities.		
	Explain the law of independent assortment, using a suitable example.	G12BiU08P02	Law of Independent Assortment
	Describe the exceptions to the Mendel's laws of inheritance.	G12BiU08P05	Exceptions to Mendelian Inheritance and Blood Group System
	Explain incomplete dominance and exemplify it through the inheritance of flower color in 4 O' clock plant.		
	Differentiate between incomplete dominance and co-dominance.		
	Describe multiple alleles and state the alleles responsible for the trait of ABO blood groups.		
	Name the various human blood group systems.		
	Associate multiple alleles with the ABO blood group system.		
	Associate the positive and negative blood groups with the presence and absence of Rh factor.	G12BiU08P06	Rh Blood Group System
	Justify why Rh incompatibility could be a danger to the developing foetus and mother.		

	Explain Erythroblastosis foetalis in the light of antigen-antibody reaction.		
	Suggest measures to counter the problem of Erythroblastosis foetalis before it occurs.		
	Explain the case where two alleles have equal dominance and through the genetics of human blood group of AB.	G12BiU08P07	Blood Group Systems, Epistasis, and Polygenic Inheritance
	Investigate the reasons for O-ve individual as the Universal donor and AB +ve as the Universal recipient.		
	Describe the occurrence of some other blood group systems.		
	Explain the terms; polygenic and epistasis.		
	Describe polygenic inheritance, using suitable examples from plants (grain color in wheat) and animals (skin color in man).		
	List at least five polygenic traits discovered in humans.		
	Relate polygenic inheritance with epistasis.		
	Explain the XX-XY mechanism of sex determination in Drosophila and mammals.	G12BiU08P08	Sex Determination and Sex Linkage
	Describe the XX-XO and ZZ-ZW sex determination systems and evaluate by studying the karyotype.		
	Identify the difference between homogametic and heterogametic conditions in the karyotype of male and female humans.		
	Identify male and female individuals from the karyotype of Drosophila and man.		
	Describe the concept of sex-linkage.		
	Explain the inheritance of sex-linked traits (eye color) in Drosophila.		
	Name some of the sex-linked disorders of man and drosophila.		
	Describe the sex-linked inheritance of male characters due to Y-chromosome and the effect of Hollandric genes.		
	Describe the X- linked disorders with reference to the patterns of inheritance.	G12BiU08P10	Sex Linked Disorders in Humans

	Critically analyze the inheritance of Haemophilia, colour blindness and muscular dystrophy.		
	Describe the terms gene linkage and crossing over.	G12BiU08P11	Gene Linkage and Crossing Over
CHROMOSOME AND DNA	Critically analyze the history of chromosomal theory with reference to Correns' work.	G12BiU09P01	Chromosomal Theory of Inheritance
	Critically analyze the experiments of T. H. Morgan in support of the above-mentioned theory.		
	Annotate the detailed structure of a chromosome.	G12BiU09P02	Morphology of Chromosome
	Describe the concept of gene and gene locus.	G12BiU09P03	Concept of Gene
	Explain the concept of alleles as the alternative forms of a gene.		
	Narrate the experimental work of Griffith and Hershey-Chase, which proved that DNA is the hereditary material.	G12BiU09P04	DNA as the Hereditary Material and Models of DNA Replication
	Describe the three models proposed about the mechanism of DNA replication.		
	Narrate the work of Meselson and Stahl to justify the semi-conservative replication as the correct method of replication.		
	Describe the events of the process of DNA replication.	G12BiU09P05	Mechanism of DNA Replication
	Define gene and genetic code.	G12BiU09P07	Gene and Genetic Code
	Define mutation and identify various sources of mutation.	G12BiU09P11	Mutations
	Differentiate between natural and induced mutations and mutagens.		
	Justify that most mutations are harmful.		
	Rationalize that mutations might be a contributing factor towards evolution.		
	Describe the symptoms, causes and possible available treatments of some of the gene mutations. (Sickle cell anemia, Phenylketonuria)	G12BiU09P12	Chromosomal Mutation
	Describe the symptoms, causes and possible available treatments of some of the chromosomal mutations. (Down's, Klinefelter's and Turner's syndrome)		

EVOLUTION	Describe creationism and the theory of evolution as two contradictory ideas.	G12BiU10P01	Theories of Evolution
	Relate Quranic injunctions to the process of the evolution of man.		
	Explain how biogeography provides an evidences for evolution.	G12BiU10P03	Evidences of Evolution
	Describe the evidences of evolution that come from paleontology, comparative anatomy and molecular biology.		
	Differentiate between convergent and divergent evolution on the basis of inheritance of the homologous and analogous structures.		
	Describe the theories that have been put forwarded about the mechanism of evolution of eukaryotes from prokaryotes.	G12BiU10P04	Evolution From Prokaryotes to Eukaryotes
	Justify Lamarck as an early proponent of evolution.	G12BiU10P05	Lamarckism
	Describe the theory of inheritance of acquired characters, as proposed by Lamarck.		
	Outline the steps of the evolution of the giraffe, as illustrated in Lamarckism.		
	State the drawbacks in Lamarckism.		
	Explain the theory of natural selection as proposed by Darwin.	G12BiU10P06	Theory of Natural Selection
	Describe the assumptions of the Hardy-Weinberg theorem and relate these to the factors that change the allelic frequencies of the population.	G12BiU10P07	Hardy Weinberg Theorem
MAN AND HIS ENVIRONMENT	Define the concept of speciation and explain the mechanisms of speciation (allopatric, parapatric and sympatric speciation).	G12BiU10P08	Speciation
	Define biogeochemical cycles and locate the primary reservoirs of the chemicals in these cycles.	G12BiU11P01	Biogeochemical Cycle and Water Cycle
	Describe water cycle in detail.		
	Define the terms aquifers and water table.		
	Describe nitrogen cycle in detail.	G12BiU11P02	Nitrogen Cycle
	Define the terms of nitrogen-fixation, nitrification, de-nitrification and ammonification.		

	Explain the flow of energy in successive trophic levels.	G12BiU11P04	The Flow of Energy
	Interpret the pyramids of number, biomass and energy.	G12BiU11P05	Ecological Pyramids
	Define ecological succession as the process through which ecosystems change from simple to complex.	G12BiU11P06	Ecological Succession
	Describe primary and secondary succession.		
	Differentiate between xerarch and hydrarch succession.		
	Explain the xerarch succession on a bare rock starting from the small pockets of lichens to the vegetations of flowering plants.		
	Describe characteristics of a population, such as growth, density, distribution, carrying capacity, minimum/viable size.	G12BiU11P09	Population Dynamics
	Distinguish between renewable and non-renewable environmental resources.	G12BiU11P10	Environmental Resources and Their Depletion
	Describe how man is responsible for the depletion of environmental resources.		
BIOTECHNOLOGY	Define gene cloning and state the steps in gene cloning.	G12BiU12P01	Recombinant DNA Technology
	Describe the techniques of gene cloning through recombinant DNA technology.		
	Explain the role of restriction endonucleases and DNA ligases in gene cloning.		
	Describe the selection and isolation of the gene of interest.		
	Explain the properties and the role of vectors in recombinant DNA technology.		
	State the steps for the integration of DNA insert into the vector.		
	Briefly state the technique applied for the selection of the vectors that take up the DNA insert.		
	Describe the procedure for the construction of genomic library.	G12BiU12P03	Genomic Library and Human Genome Project
	State the history of the human genome project admiring James Watson as its first director.		
	Describe the goals of the human genome project.		

	Predict some of the possible benefits that can be derived after the completion of the human genome project.		
	Describe the purposes and mechanism of DNA analysis.	G12BiU12P05	DNA Analysis
	State the objectives of the production of transgenic bacteria, transgenic plants and transgenic animals.	G12BiU12P09	Transgenic Bacteria, and Plants and Animals
	State the objectives of the production of transgenic bacteria, transgenic plants and transgenic animals.		
	State the role of transgenic bacteria in making biotechnology products.		
	State the role of transgenic bacteria in making biotechnology products.		
	List some of the ecological concerns surrounding transgenic bacteria.		
	List some of the ecological concerns surrounding transgenic bacteria.		
	Explain the scope and importance of biotechnology in promoting human welfare.		
	List the hazards and social/ethical implications of using gene technology in human.		
	Describe the ways in which genetic engineering improves farm animals.	G12BiU12P10	Transgenic Animals
	Describe the ways in which genetic engineering improves farm animals.		
BIOLOGY AND HUMAN WELFARE	Explain what is meant by integrated disease management	G12BiU13P01	Vaccination and Integrated Disease Management
	Describe vaccination and its importance.		
	List some common viral diseases against which vaccination are required e.g. polio, measles, influenza and hepatitis.		
	Describe the role of vaccines in preventing polio, measles, hepatitis and tetanus.	G12BiU13P02	Role of Vaccines in Preventing Diseases
	State the schedule of the vaccination against polio, measles, hepatitis and tetanus.		
	Describe animal husbandry and the role of life stock in national economy (milk, meat, eggs, wool)	G12BiU13P03	Animal Husbandry

	and other miscellaneous products).		
	List the outstanding milk producing breeds of cows and buffaloes.		
	Describe different methods adopted for plant improvements (acclimatization, selection, hybridization and back crosses etc).	G12BiU13P04	Methods Adopted for Plant Improvements and Home Gardening
	Explain home gardening and its importance.		
	Identify some seasonal vegetable and fruit plants suitable for home gardening.		
	Explain the role of microbes in household food processing, industrial production, sewage treatment and energy generation.	G12BiU13P05	Role of Microbes
EVOLUTION	Briefly describe the observations Darwin made during his voyage on HMS Beagle.	G12BiU10P02	Darwinism and Genetic Drift
	Describe the ideas of Charles Lyell, James Hutton and Thomas Malthus that contributed in the early development of Darwinism.		
	Describe the role of Alfred Wallace in motivating Darwin to publish the theory of natural selection.		
	Justify, on the grounds that both Wallace's and Darwin's papers were published in the Journal of the proceedings of the Linnaean Society, why the theory was attributed to Darwin.		
	Explain the concept of genetic drift (neutral selection).		
MAN AND HIS ENVIRONMENT	Relate the need of the nuclear power to the scarcity of fossil fuels.	G12BiU11P07	Human Impacts on the Environment
	State the problems of using nuclear power (surety of safe operation and safe disposal of the wastes).		
	Narrate the incidence when one of the four reactors of the		

	Chernobyl nuclear power plant blew up in 1986		
	Describe the causes of the increasing concentration of carbon dioxide in the world's atmosphere.		
	Correlate the increasing CO ₂ concentration with the global warming and describe its long term effects.		
	Explain the causes and effects of acid rain.		
	Describe the composition of the ozone layer and its role in protecting the life on earth.		
	State the sources of chlorofluorocarbons and their role in the depletion of ozone.		
	Explain the effects of ultraviolet radiation as a serious human health concern.		
	Describe the conventional and non-conventional energy resources.	G12BiU11P08	Conventional and Non Conventional Energy Resources
	Analyze the efforts of various government departments and NGOs to educate people for the protection of environmental resources.		
BIOTECHNOLOGY	Define following terms related to plant tissue culture; explants, callus, micro-propagation, plantlets, somatic embryogenesis, somaclonal variation.	G12BiU12P07	Tissue Culture
	Explain tissue culture and differentiate between the organ culture and cell culture.		
	Differentiate between the callus culture and suspension culture techniques.		
	Describe the anther culture, ovary culture, meristem culture and embryo culture techniques.		
	Briefly describe the techniques used for, applications and limitations of animal tissue culture.		

d. Chemistry

Unit	SLO	Package Code	Package Titles
s- and p - Block Elements	Recognize the demarcation of the Periodic Table into s block, p block, d block, and f block. (Understanding)	G12ChU01P01	Periodic Table Block Demarcation and Oxide Properties
	Describe physical properties and acid-base behavior of oxides, chlorides and hydroxides of period 3 elements. (Applying)		
	Describe how physical properties like atomic radius, ionization energy, electronegativity, electrical conductivity and melting and boiling points of elements change within a group and within a period in the Periodic Table. (Analyzing)	G12ChU01P02	Atomic Radius and Its Trend in Periodic Table
		G12ChU01P03	Electronegativity and Its Trend in Periodic Table
		G12ChU01P04	Trends in Periodic Properties
	Describe reactions of period 3 elements with water, oxygen and chlorine. (Applying)	G12ChU01P05	Reactions of Period 3 Elements Part 1
		G12ChU01P06	Reactions of Period 3 Elements Part 2
	Describe reactions of Group I elements with water, oxygen and chlorine. (Applying)	G12ChU01P10	Reactions of Group IA and IIA Elements
	Describe reactions of Group II elements with water, oxygen and nitrogen. (Applying)		
	Explain effect of heat on nitrates, carbonates and hydrogen carbonates of Group I elements. (Applying)	G12ChU01P12	Anomalous Nature of Beryllium
	Discuss the trend in solubility of the hydroxides, sulphates and carbonates of Group II elements. (Analyzing)		
	Discuss the trends in thermal stability of the nitrates and carbonates of Group II elements. (Analyzing)		
	Differentiate beryllium from other members of its group. (Analyzing)		
	Explain the relative behaviour of halogens as oxidizing agents and reducing agents. (Applying)	G12ChU01P14	Strength of Halogen as Oxidizing Agent

		G12ChU01P15	Strength of Halogen as a Reducing Agent
	Compare the acidity of hydrogen halides. (Analyzing)		
d and f - Block Elements (Transition Elements)	Describe electronic structures of elements and ions of d-block elements. (Applying)	G12ChU02P01	Electronic Structures of Transition Elements
	Explain why the electronic configuration for chromium and copper differ from those assigned using the Aufbau principle. (Analyzing)		
	Explain shapes, origin of colors and nomenclature of coordination compounds. (Applying)	G12ChU02P04	Nomenclature of Complex Compounds
	Relate the coordination number of ions to the crystal structure of the compound of which they are a part. (Applying)	G12ChU02P05	Shape of Complex Ions, Coordination No, Color of Complexes
	Describe the reactions of potassium dichromate with oxalic acid and Mohr's salt. (Understanding)	G12ChU02P07	Potassium Dichromate and Potassium Permanganate
	Describe the reactions of potassium manganate VII with ferrous sulphate, oxalic acid and Mohr's salt. (Understanding)		
Organic Compounds	Define organic chemistry and organic compounds. (Remembering)	G12ChU03P01	Introduction to Organic Chemistry
	Explain that organic compounds are also synthesized in the lab. (Understanding)		
	Explain why there is such a diversity and magnitude of organic compounds. (Analyzing)	G12ChU03P02	Features and Classification of Organic Compound
	Classify organic compounds on structural basis. (Analyzing)		
	Explain the use of coal as a source of both aliphatic and aromatic hydrocarbons. (Understanding)	G12ChU03P03	Natural Sources of Organic Compounds
	Explain the use of plants as a source of organic compounds. (Understanding)		
	Define functional groups and homologous series. (Remembering)	G12ChU03P04	Functional Group
Hydrocarbons	Classify hydrocarbons as aliphatic and aromatic. (Understanding)	G12ChU04P01	Introduction and Classification of Hydrocarbon
	Describe nomenclature of alkanes and cycloalkanes. (Understanding)	G12ChU04P02	Nomenclature of Alkanes

	Explain the concept of conjugation in alkenes having alternate double bonds. (Understanding)		
	Define hemolytic and heterolytic fission, free radical initiation, propagation and termination. (Remembering)	G12ChU04P06	Radical Substitution Reaction
	Describe the mechanism of free radical substitution in alkanes exemplified by methane and ethane. (Understanding)		
	Explain the nomenclature of alkenes. (Understanding)	G12ChU04P09	Nomenclature of Alkene and Alkyne
	Use the IUPAC naming system for alkenes. (Applying)		
	Use the IUPAC naming system for alkynes. (Applying)		
	Explain shape of ethene molecule in terms of sigma and pi C-C bonds. (Understanding)	G12ChU04P10	Structure of Ethene
	Describe the structure and reactivity of alkenes as exemplified by ethene. (Applying)		
	Define and explain with suitable examples the terms isomerism, stereoisomerism and structural isomerism. (Remembering)	G12ChU04P11	Isomerism and Its Types
	Explain dehydration of alcohols and dehydrohalogenation of RX for the preparation of ethene. (Understanding)	G12ChU04P12	Preparation of Ethene
	Describe the chemistry of alkenes by the following reactions of ethene: Hydrogenation, hydrohalogenation, hydration, halogenation, halohydration, epoxidation, ozonolysis, polymerization. (Understanding)	G12ChU04P13	Reactions of Ethene Part 1
		G12ChU04P14	Reactions of Ethene Part 2
	Explain the shape of benzene molecule (molecular orbital aspect). (Understanding)	G12ChU04P15	Shape of Benzene
	Describe what is meant by the term delocalized electrons in the context of the benzene ring. (Understanding)	G12ChU04P16	Benzene and Its Comparison With Alkane, Alkene, and Alkyne
	Define resonance, resonance energy and relative stability. (Understanding)		

	Compare the reactivity of benzene with alkanes and alkenes. (Applying)		
	Describe addition reactions of benzene and methyl benzene. (Applying)		
	Describe the mechanism of electrophilic substitution in benzene. (Understanding)	G12ChU04P19	Electrophilic Substitution Reaction of Benzene
	Discuss chemistry of benzene and methyl benzene by nitration, sulphonation, halogenation, Friedel Craft's alkylation and acylation. (Applying)	G12ChU04P20	Halogenation of Benzene
		G12ChU04P21	Nitration of Benzene
		G12ChU04P22	Friedel Craft Alkylation and Acylation Part 1
		G12ChU04P23	Friedel Craft Alkylation and Acylation Part 2
	Apply the knowledge of positions of substituents in the electrophilic substitution of benzene. (Applying)	G12ChU04P25	Orientation of Benzene Part 1
		G12ChU04P26	Orientation of Benzene Part 2
	Discuss the shape of alkynes in terms of sigma and pi C-C bonds. (Applying)	G12ChU04P27	Shape of Ethyne
	Describe the preparation of alkynes using elimination reactions. (Applying)	G12ChU04P28	Preparation of Ethyne and Reactions of Alkenes
	Describe acidity of alkynes. (Understanding)	G12ChU04P29	Reactions of Ethyne Part 1
	Discuss chemistry of alkynes by hydrogenation, hydrohalogenation, hydration, bromination, ozonolysis, and reaction with metals. (Understanding)	G12ChU04P30	Reactions of Ethyne Part 2
Alkyl halides and amines	Name alkyl halides using IUPAC system. (Applying)	G12ChU05P01	Nomenclature of Alkyl Halides
	Discuss the structure and reactivity of RX. (Applying)	G12ChU05P02	Classification and Preparation of Alkyl Halide
	Describe the preparation of RX by the reaction of alcohols with HX, SOCl ₂ and PX ₃ and by radical halogenation of alkanes. (Applying)		
	Describe the mechanism and types of nucleophilic substitution reactions. (Understanding)	G12ChU05P03	Nucleophilic Substitution Reaction
		G12ChU05P04	SN ₂ Reactions

	Describe the mechanism and types of elimination reactions. (Understanding)	G12ChU05P05	Elimination Reactions
		G12ChU05P06	E2 Reactions
		G12ChU05P07	E1 Reactions
	Discuss the chemistry of Grignard's reagent with the addition of aldehydes, ketones, esters, and carbon dioxide.	G12ChU05P08	Reactions of Grignard Reagent
	Describe the preparation and reactivity of Grignard's Reagents. (Applying) Discuss the chemistry of Grignard's reagent by the addition of aldehydes, ketones, esters, and carbon dioxide. (Understanding)	G12ChU05P09	Preparation and Reaction of Grignard Reagent
	Discuss nomenclature, structure and basicity of amines. (Applying)	G12ChU05P10	Amines
	Describe the preparation of amines by alkylation of ammonia to RX and reduction of nitriles, nitro and amide functional groups. (Applying)		
	Discuss reactivity of amines. (Applying)	G12ChU05P11	Reactions of Amines
	Describe isomerism in alkyl halides and amines. (Understanding)	G12ChU05P12	Isomerism in Alkyl Halides and Amines
Alcohols, Phenols and Ethers	Explain nomenclature, structure and acidity of alcohols as exemplified by ethanol. (Understanding)	G12ChU06P01	Alcohol and Its Nomenclature
	Describe the preparation of alcohols by reduction of aldehydes, ketones, carboxylic acids and esters. (Applying)	G12ChU06P02	Preparation of Alcohols
	Explain reactivity of alcohols. (Understanding)	G12ChU06P03	Reactions of Alcohols Part 1
	Describe isomerism in alcohols and phenols. (Understanding)		
	Describe the chemistry of alcohols by preparation of ethers and esters, oxidative cleavage of 1, 2-diols. (Applying)	G12ChU06P04	Reactions of Alcohols Part 2
	Discuss thiols (RSH). (Understanding)	G12ChU06P05	Thiols
	Identify ethers from their formula. (Understanding)		
	Explain the nomenclature, structure, and acidity of phenols. (Applying) Differentiate between alcohol and phenol. (Understanding)	G12ChU06P06	Phenol and Its Nomenclature

	Discuss the reactivity of phenol and their chemistry by electrophilic aromatic substitution, reaction with Na metal and oxidation. (Applying)	G12ChU06P07	Reactions of Phenol
	Describe the preparation of phenol from benzene sulphonic acid, chlorobenzene, acidic oxidation of cumene and hydrolysis of diazonium salts. (Applying)	G12ChU06P08	Preparation of Phenol
	Differentiate between alcohols and phenols. (Understanding)	G12ChU06P06-A	Acidic Properties of Phenol and Alcohol vs Phenol Comparison
	Describe isomerism in alcohols and phenols. (Understanding)		
Carbonyl Compounds I: Aldehydes and Ketones	Explain nomenclature and structure of aldehydes and ketones. (Applying)	G12ChU07P01	Nomenclature of Aldehyde and Ketone
	Discuss the preparation of aldehydes and ketones by ozonolysis of alkenes, hydration of alkynes, oxidation of alcohols and Friedel Craft's acylation of aromatics. (Applying)	G12ChU07P02	Preparation of Aldehydes and Ketones
	Describe reactivity of aldehydes and ketones and their comparison. (Analyzing)	G12ChU07P03	Isomerism in Aldehyde and Ketone
	Describe isomerism in aldehydes and ketones. (Understanding)		
	Describe oxidation reactions of aldehydes and ketones. (Applying)	G12ChU07P04	Oxidation Reactions
	Describe acid and base catalysed nucleophilic addition reactions of aldehydes and ketones. (Applying)	G12ChU07P05	Acid Catalysed Reactions of Carbonyl Compounds
		G12ChU07P06	Base Catalysed Addition Reactions of Carbonyl Compounds
		G12ChU07P06	Carbonyl Compound Addition and Reduction Reactions
	Discuss the chemistry of aldehydes and ketones by their reduction to hydrocarbons, alcohols, by using carbon nucleophiles, nitrogen nucleophiles and oxygen nucleophiles. (Applying)		

Carbonyl Compounds 2: Carboxylic Acids and Functional Derivatives	Describe preparation of carboxylic acids by carbonation of Grignard's Reagent, hydrolysis of nitriles, oxidation of primary alcohols, oxidation of aldehydes and oxidation of alkyl benzenes. (Applying)	G12ChU08P01	Preparation of Carboxylic Acid
	Discuss reactivity of carboxylic acids. (Applying) Describe the chemistry of carboxylic acids by conversion to carboxylic acid derivatives: acyl halides, acid anhydrides, esters, amides and reactions involving inter conversion of these. (Analyzing)	G12ChU08P02	Reactions of Carboxylic Acids
	Describe the chemistry of carboxylic acids by conversion to carboxylic acid derivatives: acyl halides, acid anhydrides, esters, amides and reactions involving inter conversion of these. (Analyzing)	G12ChU08P03	Chemistry of Carboxylic Acids
	Describe isomerism in carboxylic acids. (Understanding)		
	Describe reactions of carboxylic acid derivatives. (Applying)	G12ChU08P04	Reactions of Carboxylic Acids Part 1
		G12ChU08P05	Reactions of Carboxylic Acids Part 2
Biochemistry	Explain the basis of classification and structure-Function relationship of Carbohydrates (understanding)	G12ChU09P01	Carbohydrates
	Explain the role of various Carbohydrates in health and diseases(Understanding)		
	Identify the nutritional importance and their role as energy storage (Applying)		
	Explain the basis of classification and structure-function relationship of proteins (Understanding)	G12ChU09P02	Protein and Enzyme
	Describe the role of various proteins in maintaining body functions and their nutritional importance (Applying)		
	Describe the role of enzyme as biocatalyst and relate this role to various functions such as digestion of food (Applying)		
	Identify factors that affect enzyme activity such as effect of temperature and pH.		

	Explain the role of inhibitors of enzyme catalyzed reactions (Understanding)	G12ChU09P03	Inhibitors
	Identify the nutritional and Biological importance of lipids (Applying)	G12ChU09P06	Lipids
	Describe the basis of classification and structure-Function relationship of Lipids (Applying)		
	Identify the structural components of DNA and RNA (Applying)	G12ChU09P04	DNA and RNA
	Recognize the structural differences between DNA polymer (double strand) and RNA (single strand).		
	Relate DNA sequences to its function as storage of genetic information (Applying)		
	Relate RNA sequence (transcript) to its role in transfer of information to protein (Translation) (Applying)	G12ChU09P05	RNA and Its Types
Industrial Chemistry	Describe the chemical processes of addition and condensation polymerization. (Understanding)	G12ChU10P02	Addition and Condensation Polymerization
	Interpret difference between petrochemical and chemicals derived from them.(Understanding)	G12ChU10P03	Petroleum and Its Fractional Distillation
	Describe the fractional distillation and refining of Petroleum (Understanding)		
	Describe the basic building block processes in Petrochemical technology.(Understanding)		
	List the various raw materials for Petrochemical industry. (Analyzing)		
	Identify the important fractions. (Analyzing)		
	Describe the Petrochemical process technology. (Understanding)		
	List some major petrochemicals. (Applying)		
	List some major products of the petrochemicals industry, together with their uses. (Applying)		
	Discuss types and applications of hair dyes. (Applying)	G12ChU10P04	Fiber, Its Types, and Hair Dyes
	Describe the formation and uses of PVC and Nylon. (Applying)		
Environmental Chemistry	Recognize various chemical reactions occurring in the atmosphere. (Understanding)	G12ChU11P01	Chemistry of Troposphere and Greenhouse Effect
	Recognize that the release of CO _x , SO _x , NO _x , VOCs are associated with the		

	combustion of hydrocarbon based fuels. (Applying)		
	Outline problems associated with release of pollutants including acid rain and the formation by free radical reactions of hazardous inorganic and organic compounds e.g., PAN. (Analyzing)		
	Describe causes and impacts of urban smog. (Analyzing)		
	Explain greenhouse effect and global warming as resulting in climate change. (Analyzing)		
	Explain the build up to and recognize the adverse effects of ozone in the troposphere. (Applying)		
	Describe the role of CFCs in destroying ozone in the stratosphere. (Applying)	G12ChU11P02	Chemistry of Stratosphere and Water Pollution
	Describe the role of ozone in the stratosphere in reducing the intensity of harmful UV radiation reaching the earth. (Understanding)		
	List possible alternatives to the use of CFCs. (Applying)		
	Recognize and describe various water pollutants. (Applying)		
	Explain the various parameters of water analysis. (Applying)		
Analytical Chemistry	Compare the classical method of analysis with modern methods. (Understanding)	G12ChU12P01	Analytical Chemistry and Mass Spectrometer
	Discuss the procedure of combustion analysis. (Understanding)		
	Explain instrumentation and working of MS. (Applying)		
	Outline the use of MS in determination of relative isotopic masses and isotopic abundance. (Applying)		
	Define and explain atomic emission and atomic absorption spectrum. (Understanding)	G12ChU12P02	Atomic Emission and Atomic Absorption Spectrum

e. Physics

Unit	SLO	Package Code	Package Name
Thermodynamics	calculate work done by a thermodynamic system during a volume change.	G12PhU01P01	Thermodynamic System
	describe the first law of thermodynamics expressed in terms of the change in internal energy, the heating of the system and work done on the system.		
	explain that first law of thermodynamics expresses the conservation of energy.		
	define the terms, specific heat and molar specific heats of a gas.	G12PhU01P02	Specific Heat, Reversible and Irreversible Process
	apply first law of thermodynamics to derive $C_p - C_v = R$.		
	describe the concept of reversible and irreversible processes.		
	state and explain second law of thermodynamics.	G12PhU01P03	Carnot Engine and 2nd Law of Thermodynamics
	explain the working principle of Carnot's engine		
	state the working principle of heat engine.	G12PhU01P04	Heat Engine and Entropy
	describe that change in entropy is positive when heat is added and negative when heat is removed from the system.		
	explain that increase in temperature increases the disorder of the system.		
	explain that increase in entropy means degradation of energy.		
Electrostatics	explain that energy is degraded during all natural processes.		
	state Coulomb's law and explain that force between two point charges is reduced in a medium other than free space using Coulomb's law.	G12PhU02P05	Coulomb's Law and Electric Field Strength
	define electric field strength as force per unit positive charge .		
	solve problems and analyze information using $E = F/q$.	G12PhU02P06	Numerical Problems and Gauss's Law
	solve problems involving the use of the expression . $E = 1/4\pi\epsilon_0 q/r^2$		

	solve problems by using the expression $E = V/d$.		
	state and explain Gauss's law.		
	calculate the magnitude and direction of the electric field at a point due to two charges with the same or opposite signs.	G12PhU02P07	Concept of Electric Field
	sketch the electric field lines for two point charges of equal magnitude with same or opposite signs.		
	define and explain electric flux.		
	describe electric flux through a surface enclosing a charge.		
	derive the expression $E = 1/4\pi\epsilon_0 q/r^2$ for the magnitude of the electric field at a distance 'r' from a point charge 'q'.		
	describe and draw the electric field due to an infinite size conducting plate of positive or negative charge.		
	sketch the electric field produced by a hollow spherical charged conductor.	G12PhU02P08	Electric Flux and Electric Field Intensity
	sketch the electric field between and near the edges of two infinite size oppositely charged parallel plates.		
	define electric potential at a point in terms of the work done in bringing unit positive charge from infinity to that point.	G12PhU02P09	Introduction to Electric Potential and Potential Difference
	define the unit of potential.		
	calculate the potential in the field of a point charge using the equation $V = 1/4\pi\epsilon_0 q/r$.		
	describe that the electric field at a point is given by the negative of potential gradient at that point.		
	derive an expression for electric potential at a point due to a point charge.	G12PhU02P10	Derivation of Electric Potential Expression
	define and become familiar with the use of electron volt.	G12PhU02P11	Electron Volt and Capacitors
	describe the functions of capacitors in simple circuits.		
	explain polarization of dielectric of a capacitor.	G12PhU02P12	Capacitors' Combination and Capacitance
	define capacitance and the farad and solve problems by using $C=Q/V$.		

	solve problems using formula for capacitors in series and in parallel.		
	demonstrate charging and discharging of a capacitor through a resistance.	G12PhU02P13	Working Principle of Capacitor
	prove that energy stored in a capacitor is $W = \frac{1}{2}QV$ and hence $W = \frac{1}{2}CV^2$.		
Current Electricity	describe the concept of steady current.	G12PhU03P14	Electric Current and Ohm's law
	state Ohm's law.		
	State ohm's law	G12PhU03P15	Ohm's Law and Resistivity
	define resistivity and explain its dependence upon temperature.		
	define conductance and conductivity of conductor.	G12PhU03P16	Conductance, Rheostat, Thermistor, EMF and Potential Difference
	describe the working of rheostat in the potential divider circuit.		
	state the characteristics of a thermistor and its use to measure low temperatures.		
	explain variation of thermoelectric e.m.f. with temperature.		
	distinguish between e.m.f and p.d. using the energy considerations.		
	explain the internal resistance of sources and its consequences for external circuits.		
	describe some sources of e.m.f.	G12PhU03P17	Maximum Power Output and Kirchhoff's Rules
	describe the conditions for maximum power transfer.		
	apply Kirchhoff's first law as conservation of charge to solve problem.		
	apply Kirchhoff's second law as conservation of energy to solve problem.	G12PhU03P18	Wheatstone Bridge and Potentiometer
	describe what is a Wheatstone bridge and how it is used to find unknown resistance.		
	describe the function of potentiometer to measure and compare potentials without drawing any current from the circuit.		
Electromagnetism	explain that magnetic field is an example of a field of force produced either by current-carrying conductors or by permanent magnets.	G12PhU04P19	Magnetic Field

	describe magnetic effect of current.		
	describe and sketch field lines pattern due to a long straight wire.		
	explain that a force might act on a current-carrying conductor placed in a magnetic field.		
	Investigate the factors affecting the force on a current carrying conductor in a magnetic field.		
	solve problems involving the use of $F = BIL \sin \theta$.		
	define magnetic flux density and its units.	G12PhU04P20	Magnetic Flux and Galvanometer
	describe the concept of magnetic flux (Φ) as scalar product of magnetic field (B) and area (A) using the relation $\Phi B = B \perp A = B.A$.		
	explain how a given galvanometer can be converted into a voltmeter or ammeter of a specified range.		
	state Ampere's law.	G12PhU04P21	Ampere's Law and Motion of Charged Particle
	apply Ampere's law to find magnetic flux density around a wire and inside a solenoid.		
	describe quantitatively the path followed by a charged particle shot into a magnetic field in a direction perpendicular to the field.		
	explain that a force may act on a charged particle in a uniform magnetic field.		
	describe a method to measure the e/m of an electron by applying magnetic field and electric field on a beam of electrons.	G12PhU04P22	e/m of an Electron and Torque on Current Carrying Coil
	predict the turning effect on a current carrying coil in a magnetic field and use this principle to understand the construction and working of a galvanometer.		
	describe the use of avometer / multimeter (analogue and digital).	G12PhU04P23	AVO Meter
Electromagnetic Induction	describe the production of electricity by magnetism.	G12PhU05P24	Induced EMF and Induced Current
	explain that induced emf's can be generated in two ways. (i) by relative		

	movement (the generator effect). (ii) by changing a magnetic field (the transformer effect).		
	infer the factors affecting the magnitude of the induced emf.		
	explain the production of back emf in electric motors.	G12PhU05P25	Motional EMF and Mutual Induction
	explain what is meant by motional emf. Given a rod or wire moving through a magnetic field in a simple way, compute the potential difference across its ends.		
	state Faraday's law of electromagnetic induction.	G12PhU05P26	Faraday's Law
	apply Faraday's law of electromagnetic induction and Lenz's law to solve problems.		
	describe the construction of a transformer and explain how it works.	G12PhU05P27	Transformer and Alternating Current Generator
	identify the relationship between the ratio of the number of turns in the primary and secondary coils and the ratio of primary to secondary voltages.		
	describe how set-up and step-down transformers can be used to ensure efficient transfer of electricity along cables.		
	describe the main components of an A.C generator and explain how it works.		
	account for Lenz's law to predict the direction of an induced current and relate to the principle of conservation of energy.	G12PhU05P28	Lenz's Law and Energy Stored in an Inductor
	describe the main features of an A.C electric motor and the role of each feature.		
Alternating Current	describe the terms time period, frequency, instantaneous peak value and root mean square value of an alternating current and voltage.	G12PhU06P29	Alternating Current and Its Phases
	represent a sinusoidally alternating current or voltage by an equation of the form $x = x_0 \sin \omega t$.		
	describe the phase of A.C and how phase lags and leads in A.C Circuits.		
		G12PhU06P30	

	explain the flow of A.C through resistors, capacitors and inductors. / apply the knowledge to calculate the reactance of capacitors and inductors.		AC Through Resistor, Capacitor, and Inductor
	construct phasor diagrams and carry out calculations on circuits including resistive and reactive components in series.	G12PhU06P31	Power in AC Circuits, RC and RL Series Circuit
	solve the problems using the formulae of A.C Power, explain resonance in an A.C circuit and carry out calculations using the resonant frequency formulae.		
	describe the qualitative treatment of Maxwell's equations and production of electromagnetic waves.	G12PhU06P32	Electromagnetic Waves, Impedance and Phase Supply
	describe impedance as vector summation of resistances and reactance.		
	identify inductors as important components of A.C circuits termed as chokes (devices which present a high resistance to alternating current).		
	solve the problems using the formulae of A.C Power, explain resonance in an A.C circuit and carry out calculations using the resonant frequency formulae.		
Physics of Solids	define and use the terms Young's modulus, bulk modulus and shear modulus.	G12PhU07P33	Module of Elasticity, Properties of Solids and Strains
	become familiar of ultimate tensile stress, elastic deformation and plastic deformation of a material.		
	describe that deformation in solids is caused by a force and that in one dimension, the deformation can be tensile or compressive.	G12PhU07P34	Elastic Limit, Yield Length, Solids and Energy Bands
	classify hard and soft ferromagnetic substances.		
	describe the idea about energy bands in solids.		
	classify insulators, conductors, semiconductors on the basis of energy bands.		
	become familiar with the behavior of superconductors and their potential uses.		

	describe the concepts of magnetic domains in a material.	G12PhU07P35	Ferromagnetic Substance and Hysteresis Loop
	describe hysteresis loss.		
	synthesize from hysteresis loop how magnetic field strength varies with magnetizing current.		
Electronics	distinguish between P & N type substances.	G12PhU08P36	PN Junction and Transistors
	explain the concept of holes and electrons in semiconductors.		
	explain how electrons and holes flow across a junction.		
	describe a PN junction and discuss its forward and reverse biasing.		
	distinguish PNP & NPN transistors.		
	describe the operations of transistors.	G12PhU08P37	Rectification
	define rectification and describe the use of diodes for half and full wave rectifications.		
	deduce current equation and apply it to solve problems on transistors.	G12PhU08P38	Current Flow in a Transistor and Transistors as Amplifier
	explain the use of transistors as a switch and an amplifier.		
Dawn of the Modern Physics	describe the concept of black body radiation.	G12PhU09P39	Black Body Radiation and Planck's Theory
	describe how energy is distributed over the wavelength range for several values of source temperature.		
	describe the Planck's hypothesis that radiation emitted and absorbed by the walls of a black body cavity is quantized.		
	describe the phenomenon of photoelectric effect.	G12PhU09P40	Photoelectric Effect
	solve problems and analyze information using: $E = hf$ and $c = f\lambda$.		
	describe the relationship between Photoelectric effect and Quantum Theory	G12PhU09P41	Uncertainty Principle, Photoelectric Effect and Quantum Th.
	identify data sources, gather, process and present information to summarize the use of the photoelectric effect in solar cells & photocells		
	describe uncertainty principle		
	describe the confirmation of de Broglie's proposal by Davisson and	G12PhU09P42	

	Germer experiment in which the diffraction of electrons by the surface layers of a crystal lattice was observed.		Wave Particle, Compton Effect, Davisson and Germer Experiment
	describe the impact of de Broglie's proposal that any kind of particle has both wave and particle properties.		
	describe Compton effect qualitatively.		
	elaborate the particle nature of electromagnetic radiation.		
	explain the particle model of light in terms of photons with particular energy and frequency.		
	explain the phenomena of pair production and pair annihilation.	G12PhU09P43	Pair Production, Pair Annihilation and Electron Microscope
	explain how the very short wavelength of electrons, and the ability to use electrons and magnetic fields to focus them, allows electron microscope to achieve very high resolution.		
Atomic Spectra	explain the terms spontaneous emission, stimulated emission, meta stable states, population inversion and laser action.	G12PhU10P44	Laser
	Solve problems and analyze information using.		
	analyze the significance of the hydrogen spectrum in the development of Bohr's model of the atom.	G12PhU10P45	Bohr Model, Inner Shell Transitions and Helium Neon Laser
	explain hydrogen atom in terms of energy levels on the basis of Bohr Model.		
	determine the ionization energy and various excitation energies of an atom using an energy level diagram.		
	understand that inner shell transitions in heavy elements result into emission of characteristic X-rays.		
	describe the structure and purpose of the main components of a He-Ne gas laser.		
Nuclear Physics	explain the relevance of binding energy per nucleon to nuclear fusion and to nuclear fission.	G12PhU11P46	Binding Energy, Radioactivity and Nuclear Reactions

	identify that some nuclei are unstable, give out radiation to get rid of excess energy and are said to be radioactive.		
	determine the release of energy from different nuclear reactions.		
	explain that atomic number and mass number conserve in nuclear reactions. / describe the phenomena of nuclear fission and fusion./ describe the fission chain reaction.	G12PhU11P47	Nuclear Reactions, Nuclear Fission and Nuclear Fusion
	describe energy and mass conservation in simple reactions and in radioactive decay.	G12PhU11P48	Mass Defect, Binding Energy and Radiation with Matter
	describe the interaction of nuclear radiation with matter.		
	describe the function of various components of a nuclear reactor.	G12PhU11P49	Nuclear Reactors and Its Types
	describe the use of Geiger Muller counter and solid state detectors to detect the radiations.	G12PhU11P50	Geiger Muller Tube and Building Blocks of Matter
	describe the basic forces of nature.		
	describe the key features and components of the standard model of matter including hadrons, leptons and quarks.		
	describe a simple model for the atom to include protons, neutrons and electrons.		
Thermodynamics	explain that the efficiency of a Carnot engine is independent of the nature of the working substance and depends on the temperatures of hot and cold reservoirs.	G12PhU01P51	Carnot Engine and Refrigerator as Reverse Heat Engine
	describe that refrigerator is a heat engine operating in reverse as that of an ideal heat engine.		
	derive an expression for the coefficient of performance of a refrigerator.	G12PhU01P52	Performance of a Refrigerator and Thermodynamic System
	identify that system tend to become less orderly over time.		
	describe the concept of reversible and irreversible processes.		
Electrostatics	describe the concept of an electric field as an example of a field of force.	G12PhU02P53	Electron Dipole, Electric Field and Gauss's Law
	describe the concept of electric dipole.		

	state and explain Gauss's law.		
Electromagnetic Induction	explain the production of eddy currents and identify their magnetic and heating effects.	G12PhU05P54	Motional EMF, Eddy Current and Laminated Iron Cores
	explain the need for laminated iron cores in electric motors, generators and transformers.		
	explain the production of back emf in electric motors.		
Dawn of the Modern Physics	distinguish between inertial and non-inertial frames of reference .	G12PhU09P55	Speed of Light, Theory of Relativity and Frames of Reference
	describe the significance of Einstein's assumption of the constancy of the speed of light.		
	explain qualitatively and quantitatively the consequence of special relativity in relation to: – the relativity of simultaneity – the equivalence between mass and energy – length contraction - time dilation - mass increase		
	identify that if c is constant then space and time become relative.	G12PhU09P56	Theory of Relativity, Implications of Space Travel and Time
	explain qualitatively and quantitatively the consequence of special relativity in relation to: – the relativity of simultaneity – the equivalence between mass and energy – length contraction - time dilation - mass increase		
	explain the implications of mass increase, time dilation and length contraction for space travel.		

f. Urdu

Unit	SLO	Package Code	Package Name
شہرت عام بقائے دوام کا دربار	بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	G12UrU01P01	شہرت عام بقائے دوام کا دربار حصہ 1
	سن کر طویل کلام کے اہم نکات، جملے، مصرعے یا اشعار یاد رکھ سکے۔	G12UrU01P02	شہرت عام بقائے دوام کا دربار حصہ 2
	استحسان اور تنقیدی گفتگو سن کر سمجھ سکے اور اپنی رائے دے پائے۔	G12UrU01P03	شہرت عام بقائے دوام کا دربار حصہ 3
	طویل کلام کو اپنے حافظے سے تسلسل کے ساتھ دہرا سکے۔	G12UrU01P04	شہرت عام بقائے دوام کا دربار حصہ 4
	مجازی، ادبی اور اصطلاحی محضر میں گفتگو کر سکے۔	G12UrU01P05	شہرت عام بقائے دوام کا دربار حصہ 5
	کسی تحریر پر استحسانی/تنقیدی گفتگو کر سکے۔	G12UrU01P06	شہرت عام بقائے دوام کا دربار حصہ 6
	اخبارات اور جرائد میں خبروں، فیچروں، اداروں، رپورٹوں اشتہاروں اور خطوط بنا م مدیر کو ان میں پوشیدہ معنی اور معنی پر اثر انداز ہونے کے حوالے سے سمجھ کر پڑھ سکے۔	G12UrU01P07	شہرت عام بقائے دوام کا دربار حصہ 7
مسلمانوں کا قدیم طرز تعلیم		G12UrU02P08	مسلمانوں کا قدیم طرز تعلیم حصہ 1
	نئے متن کو سمجھ کر استحسانی سوالات کا جواب تحریر کر سکے اور عبارت کا عنوان تجویز کر سکے	G12UrU02P09	مسلمانوں کا قدیم طرز تعلیم حصہ 2
	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سکے۔ فنی و فکری تجزیہ کر کے اپنی رائے دے سکے اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سکے۔	G12UrU02P10	مسلمانوں کا قدیم طرز تعلیم حصہ 3
	کسی علمی مضمون میں استعمال کیے گئے الفاظ و اصطلاحات کو سمجھ سکے اور نفس مضمون کا ادراک کر سکے۔	G12UrU02P11	مسلمانوں کا قدیم طرز تعلیم حصہ 4
سقراط	ادب پڑھ کر تخلیقی صلاحیتیں پیدا کر سکے۔ مطبوعہ و غیر مطبوعہ مواد لکھ اور پڑھ سکے۔ خود بھی اس قسم کی تحریر لکھ سکے۔	G12UrU03P12	سقراط حصہ 1
	بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	G12UrU03P13	سقراط حصہ 2
فاقہ میں روزہ	بات درمیان سے سن کر بتا دینے پر قادر ہو سکے۔	G12UrU05P14	فاقہ میں روزہ حصہ 1
	مجازی، ادبی اور اصطلاحی محضر میں گفتگو کر سکے۔	G12UrU05P15	فاقہ میں روزہ حصہ 2
	اخبارات اور جرائد میں خبروں، فیچروں، اداروں، رپورٹوں اشتہاروں اور خطوط بنا م مدیر کو ان میں پوشیدہ معنی اور معنی پر اثر انداز ہونے کے حوالے سے سمجھ کر پڑھ سکے۔	G12UrU05P16	فاقہ میں روزہ حصہ 3
چند روز ایک روڈ رولر کے ساتھ	نئے متن کو سمجھ کر استحسانی سوالات کا جواب تحریر کر سکے اور عبارت کا عنوان تجویز کر سکے	G12UrU06P17	چند روز ایک روڈ رولر کے ساتھ حصہ 1

	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سکے۔ فنی و فکری تجزیہ کر کے اپنی رائے دے سکے اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سکے۔	G12UrU06P18	چند روز ایک روڈ رولر کے ساتھ حصہ 2
	کسی ادب پارے کے مرکزی خیال کو بیان کر سکے اور تشریح کر سکے نیز اہم نکات کا خلاصہ لکھ سکے۔	G12UrU06P19	چند روز ایک روڈ رولر کے ساتھ حصہ 3
	کسی ادب پارے کے حسن و قبح کا اندازہ کر سکے۔	G12UrU07P20	منظور حصہ 1
منظور	بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	G12UrU07P21	منظور حصہ 2
	احساس، جذبے اور تاثر کے حوالے سے مجازی زبان سن کر حقیقت کا ادراک کر سکے۔	G12UrU07P22	منظور حصہ 3
	کسی بھی عبارت کو رفتار مطالعہ کے ساتھ پڑھ سکے۔	G12UrU08P23	کتبہ حصہ 1
	کسی ادب پارے کے مرکزی خیال کو بیان کر سکے اور تشریح کر سکے نیز اہم نکات کا خلاصہ لکھ سکے۔	G12UrU08P24	کتبہ حصہ 2
کتبہ	کسی علمی مضمون میں استعمال کیے گئے الفاظ و اصطلاحات کو سمجھ سکے اور نفس مضمون کا ادراک کر سکے۔	G12UrU08P25	کتبہ حصہ 3
	ادب پڑھ کر تخلیقی صلاحیتیں پیدا کر سکے۔ مطبوعہ و غیر مطبوعہ مواد لکھ اور پڑھ سکے۔ خود بھی اس قسم کی تحریر لکھ سکے۔	G12UrU08P26	کتبہ حصہ 4
	بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	G12UrU09P27	مائیں حصہ 1
مائیں	بات درمیان سے سن کر بتا دینے پر قادر ہو سکے۔	G12UrU09P28	مائیں حصہ 2
	مجازی، ادبی اور اصطلاحی محضر میں گفتگو کر سکے۔	G12UrU09P29	مائیں حصہ 3
	اخبارات اور جرائد میں خبروں، فیچروں، اداروں، رپورٹوں اشتہاروں اور خطوط بنا م مدیر کو ان میں پوشیدہ معنی اور معنی پر اثر انداز ہونے کے حوالے سے سمجھ کر پڑھ سکے۔	G12UrU10P30	سیاہ فام حصہ 1
سیاہ فام	نئے متن کو سمجھ کر استحسانی سوالات کا جواب تحریر کر سکے اور عبارت کا عنوان تجویز کر سکے	G12UrU10P31	سیاہ فام حصہ 2
	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سکے۔ فنی و فکری تجزیہ کر کے اپنی رائے دے سکے اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سکے۔	G12UrU10P32	سیاہ فام حصہ 3
	کسی ادب پارے کے مرکزی خیال کو بیان کر سکے اور تشریح کر سکے نیز اہم نکات کا خلاصہ لکھ سکے۔	G12UrU11P33	محسن محلہ حصہ 1
محسن محلہ	کسی ادب پارے کے حسن و قبح کا اندازہ کر سکے۔	G12UrU11P34	محسن محلہ حصہ 2
کنڈکٹر	بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	G12UrU12P35	کنڈکٹر حصہ 1

	احساس، جذبے اور تاثر کے حوالے سے مجازی زبان سن کر حقیقت کا ادراک کر سکے۔	G12UrU12P36	کنڈکٹر حصہ 2
	کسی بھی عبارت کو رفتار مطالعہ کے ساتھ پڑھ سکے۔	G12UrU12P37	کنڈکٹر حصہ 3
	کسی ادب پارے کے مرکزی خیال کو بیان کر سکے اور تشریح کر سکے نیز اہم نکات کا خلاصہ لکھ سکے۔	G12UrU12P38	کنڈکٹر حصہ 4
ایک وصیت کی تعمیل	سن کر طویل کلام کے اہم نکات، جملے، مصرعے یا اشعار یاد رکھ سکے۔	G12UrU13P39	ایک وصیت کی تعمیل حصہ 1
	استحسان اور تنقیدی گفتگو سن کر سمجھ سکے اور اپنی رائے دے پائے۔	G12UrU13P40	ایک وصیت کی تعمیل حصہ 2
	طویل کلام کو اپنے حافظے سے تسلسل کے ساتھ دہرا سکے۔	G12UrU13P41	ایک وصیت کی تعمیل حصہ 3
	مجازی، ادبی اور اصطلاحی محضر میں گفتگو کر سکے۔	G12UrU13P42	ایک وصیت کی تعمیل حصہ 4
علامہ اقبال	کسی تحریر پر استحسانی/تنقیدی گفتگو کر سکے۔	G12UrU14P43	علامہ اقبال حصہ 1
	اخبارات اور جرائد میں خبروں، فیچروں، اداروں، رپورٹوں اشتہاروں اور خطوط بنا م مدیر کو ان میں پوشیدہ معنی اور معنی پر اثر انداز ہونے کے حوالے سے سمجھ کر پڑھ سکے۔	G12UrU14P44	علامہ اقبال حصہ 2
		G12UrU14P45	علامہ اقبال حصہ 3
اے طائر لاپہوتی	نئے متن کو سمجھ کر استحسانی سوالات کا جواب تحریر کر سکے اور عبارت کا عنوان تجویز کر سکے	G12UrU15P46	اے طائر لاپہوتی حصہ 1
	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سکے۔ فنی و فکری تجزیہ کر کے اپنی رائے دے سکے اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سکے۔	G12UrU15P47	اے طائر لاپہوتی حصہ 2
	کسی علمی مضمون میں استعمال کیے گئے الفاظ و اصطلاحات کو سمجھ سکے اور نفس مضمون کا ادراک کر سکے۔	G12UrU15P48	اے طائر لاپہوتی حصہ 3
مرید پور کا پیر	ادب پڑھ کر تخلیقی صلاحیتیں پیدا کر سکے۔ مطبوعہ و غیر مطبوعہ مواد لکھ اور پڑھ سکے۔ خود بھی اس قسم کی تحریر لکھ سکے۔	G12UrU16P49	مرید پور کا پیر حصہ 1
	بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	G12UrU16P50	مرید پور کا پیر حصہ 2
	بات درمیان سے سن کر بتا دینے پر قادر ہو سکے۔	G12UrU16P51	مرید پور کا پیر حصہ 3
حاجی اورنگ زیب خان	مجازی، ادبی اور اصطلاحی محضر میں گفتگو کر سکے۔	G12UrU17P52	حاجی اورنگ زیب خان حصہ 1
	اخبارات اور جرائد میں خبروں، فیچروں، اداروں، رپورٹوں اشتہاروں اور خطوط بنا م مدیر کو ان میں پوشیدہ معنی اور معنی پر اثر انداز ہونے کے حوالے سے سمجھ کر پڑھ سکے۔	G12UrU17P53	حاجی اورنگ زیب خان حصہ 2
	نئے متن کو سمجھ کر استحسانی سوالات کا جواب تحریر کر سکے اور عبارت کا عنوان تجویز کر سکے	G12UrU17P54	حاجی اورنگ زیب خان حصہ 3
	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سکے۔ فنی و فکری تجزیہ	G12UrU17P55	حاجی اورنگ زیب خان حصہ 4

	کر کے اپنی رائے دے سکے اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سکے۔		
	کسی ادب پارے کے مرکزی خیال کو بیان کر سکے اور تشریح کر سکے نیز اہم نکات کا خلاصہ لکھ سکے۔	G12UrU17P56	حاجی اورنگ زیب خان حصہ 5
	کسی ادب پارے کے حسن و قبح کا اندازہ کر سکے۔	G12UrU18P57	جواب شکوہ حصہ 1
	- بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	G12UrU18P58	جواب شکوہ حصہ 2
	احساس، جذبے اور تاثر کے حوالے سے مجازی زبان سن کر حقیقت کا ادراک کر سکے۔	G12UrU18P59	جواب شکوہ حصہ 3
	- کسی بھی عبارت کو رفتار مطالعہ کے ساتھ پڑھ سکے۔	G12UrU18P60	جواب شکوہ حصہ 4
	کسی ادب پارے کے مرکزی خیال کو بیان کر سکے اور تشریح کر سکے نیز اہم نکات کا خلاصہ لکھ سکے۔	G12UrU18P61	جواب شکوہ حصہ 5
	- کسی علمی مضمون میں استعمال کیے گئے الفاظ و اصطلاحات کو سمجھ سکے اور نفس مضمون کا ادراک کر سکے۔	G12UrU18P62	جواب شکوہ حصہ 6
جواب شکوہ	ادب پڑھ کر تخلیقی صلاحیتیں پیدا کر سکے۔ مطبوعہ و غیر مطبوعہ مواد لکھ اور پڑھ سکے - خود بھی اس قسم کی تحریر لکھ سکے۔	G12UrU18P63	جواب شکوہ حصہ 7
	- بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	G12UrU19P64	بڑھ چلو حصہ 1
بڑھ چلو	بات درمیان سے سن کر بتا دینے پر قادر ہو سکے۔	G12UrU19P65	بڑھ چلو حصہ 2
	مجازی، ادبی اور اصطلاحی محضر میں گفتگو کر سکے۔	G12UrU20P66	مناظر سحر حصہ 1
	اخبارات اور جرائد میں خبروں، فیچروں، اداریوں، رپورٹوں اشتہاروں اور خطوط بنا م مدیر کو ان میں پوشیدہ معنی اور معنی پر اثر انداز ہونے کے حوالے سے سمجھ کر پڑھ سکے۔	G12UrU20P67	مناظر سحر حصہ 2
	نئے متن کو سمجھ کر استحسانی سوالات کا جواب تحریر کر سکے اور عبارت کا عنوان تجویز کر سکے	G12UrU20P68	مناظر سحر حصہ 3
مناظر سحر	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سکے۔ فنی و فکری تجزیہ کر کے اپنی رائے دے سکے اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سکے۔	G12UrU20P69	مناظر سحر حصہ 4
	کسی ادب پارے کے مرکزی خیال کو بیان کر سکے اور تشریح کر سکے نیز اہم نکات کا خلاصہ لکھ سکے۔	G12UrU21P70	شکست کی آواز حصہ 1
	کسی ادب پارے کے حسن و قبح کا اندازہ کر سکے۔	G12UrU21P71	شکست کی آواز حصہ 2
	- بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	G12UrU21P72	شکست کی آواز حصہ 3
شکست کی آواز	احساس، جذبے اور تاثر کے حوالے سے مجازی زبان سن کر حقیقت کا ادراک کر سکے۔	G12UrU21P73	شکست کی آواز حصہ 4

ستارے	ادب پڑھ کر تخلیقی صلاحیتیں پیدا کر سیک۔ مطبوعہ و غیر مطبوعہ مواد لکھ اور پڑھ سیک۔ خود بھی اس قسم کی تحریر لکھ سیک۔	G12UrU22P74	ستارے حصہ 1
	بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سیک۔	G12UrU22P75	ستارے حصہ 2
نفیر عمل	بات درمیان سے سن کر بتا دینے پر قادر ہو سیک۔	G12UrU23P76	نفیر عمل حصہ 1
	مجازی، ادبی اور اصطلاحی محضر میں گفتگو کر سیک۔	G12UrU23P77	نفیر عمل حصہ 2
	اخبارات اور جرائد میں خبروں، فیچروں، اداروں، رپورٹوں اشتہاروں اور خطوط بنا م مدیر کو ان میں پوشیدہ معنی اور معنی پر اثر انداز ہونے کے حوالے سے سمجھ کر پڑھ سیک۔	G12UrU23P78	نفیر عمل حصہ 3
ہمیشہ دیر کردیتاہوں	نئے متن کو سمجھ کر استحصانی سوالات کا جواب تحریر کر سیک اور عبارت کا عنوان تجویز کر سیک	G12UrU24P79	ہمیشہ دیر کردیتاہوں حصہ 1
	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سیک۔ فنی و فکری تجزیہ کر کے اپنی رائے دے سیک اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سیک۔	G12UrU24P80	ہمیشہ دیر کردیتاہوں حصہ 2
بڑے ڈریوک ہو	کسی ادب پارے کے مرکزی خیال کو بیان کر سیک اور تشریح کر سیک نیز اہم نکات کا خلاصہ لکھ سیک۔	G12UrU25P81	بڑے ڈریوک ہو حصہ 1
	کسی ادب پارے کے حسن و قبح کا اندازہ کر سیک۔	G12UrU25P82	بڑے ڈریوک ہو حصہ 2
غزل اقبال	بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سیک۔	G12UrU26P83	غزل اقبال حصہ 1
	احساس، جذبے اور تاثر کے حوالے سے مجازی زبان سن کر حقیقت کا ادراک کر سیک۔	G12UrU26P84	غزل اقبال حصہ 2
	ادب پڑھ کر تخلیقی صلاحیتیں پیدا کر سیک۔ مطبوعہ و غیر مطبوعہ مواد لکھ اور پڑھ سیک۔ خود بھی اس قسم کی تحریر لکھ سیک۔	G12UrU26P85	غزل اقبال حصہ 3
	بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سیک۔	G12UrU26P86	غزل اقبال حصہ 4
	بات درمیان سے سن کر بتا دینے پر قادر ہو سیک۔	G12UrU26P87	غزل اقبال حصہ 5
	مجازی، ادبی اور اصطلاحی محضر میں گفتگو کر سیک۔	G12UrU27P88	غزل فیض حصہ 1
غزل فیض	اخبارات اور جرائد میں خبروں، فیچروں، اداروں، رپورٹوں اشتہاروں اور خطوط بنا م مدیر کو ان میں پوشیدہ معنی اور معنی پر اثر انداز ہونے کے حوالے سے سمجھ کر پڑھ سیک۔	G12UrU27P89	غزل فیض حصہ 2
غزل احمد ندیم قاسمی	نئے متن کو سمجھ کر استحصانی سوالات کا جواب تحریر کر سیک عبارت کا عنوان تجویز کر سیک	G12UrU28P90	غزل احمد ندیم قاسمی حصہ 1
	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سیک۔ فنی و فکری تجزیہ	G12UrU28P91	غزل احمد ندیم قاسمی حصہ 2

	کر کے اپنی رائے دے سکے اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سکے۔		
	کسی ادب پارے کے مرکزی خیال کو بیان کر سکے اور تشریح کر سکے نیز اہم نکات کا خلاصہ لکھ سکے۔	G12UrU28P92	غزل احمد ندیم قاسمی حصہ 3
	کسی ادب پارے کے حسن و قبح کا اندازہ کر سکے۔	G12UrU28P93	غزل احمد ندیم قاسمی حصہ 4
	بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	G12UrU29P94	غزل ناصر کاظمی حصہ 1
	احساس، جذبے اور تاثر کے حوالے سے مجازی زبان سن کر حقیقت کا ادراک کر سکے۔	G12UrU29P95	غزل ناصر کاظمی حصہ 2
غزل ناصر کاظمی	نئے متن کو سمجھ کر استحصانی سوالات کا جواب تحریر کر سکے اور عبارت کا عنوان تجویز کر سکے	G12UrU29P96	غزل ناصر کاظمی حصہ 3
	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سکے۔ فنی و فکری تجزیہ کر کے اپنی رائے دے سکے اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سکے۔	G12UrU30P97	غزل شکیب جلالی حصہ 1
غزل شکیب جلالی	کسی ادب پارے کے مرکزی خیال کو بیان کر سکے اور تشریح کر سکے نیز اہم نکات کا خلاصہ لکھ سکے۔	G12UrU30P98	غزل شکیب جلالی حصہ 2
	کسی ادب پارے کے حسن و قبح کا اندازہ کر سکے۔	G12UrU31P99	غزل احمد فراز حصہ 1
	بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	G12UrU31P100	غزل احمد فراز حصہ 2
غزل احمد فراز	نئے متن کو سمجھ کر استحصانی سوالات کا جواب تحریر کر سکے اور عبارت کا عنوان تجویز کر سکے	G12UrU31P101	غزل احمد فراز حصہ 3
	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سکے۔ فنی و فکری تجزیہ کر کے اپنی رائے دے سکے اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سکے۔	G12UrU32P102	غزل ظفر اقبال حصہ 1
	کسی ادب پارے کے مرکزی خیال کو بیان کر سکے اور تشریح کر سکے نیز اہم نکات کا خلاصہ لکھ سکے۔	G12UrU32P103	غزل ظفر اقبال حصہ 2
غزل ظفر اقبال	کسی ادب پارے کے حسن و قبح کا اندازہ کر سکے۔	G12UrU32P104	غزل ظفر اقبال حصہ 3
	بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	G12UrU33P105	غزل شہزاد احمد حصہ 1
غزل شہزاد احمد	کسی ادب پارے کے حسن و قبح کا اندازہ کر سکے۔	G12UrU33P106	غزل شہزاد احمد حصہ 2
معاون افعال	مرکب مصادر سے معاون افعال سیکھ سکے اور عبارت سے معاون افعال کو تلاش کر سکے۔	G12UrU34P107	معاون افعال
متعلق فعل	متعلق فعل کی تعریف و استعمال سے آگاہ ہو سکے اور عبارت سے متعلق افعال تلاش کر سکے۔	G12UrU35P108	متعلق فعل
قواعد	لغت کو اشتقاق، مشتقات، وضعی و لغوی معنی، سلینگ، بول چال، اصطلاحی معنی اور دیگر ایسی اقسام کے حوالے سے استعمال کر سکے	G12UrU36P109	تشبیہ، استعارہ

	سیکے۔ خاص علوم اور پیشوں کی اصطلاحی لغات کا استعمال سیکھ سیکے۔ 2۔ تکنیکی قواعد زبان کا ادراک کر سیکے اور لسانی انتخاب لفظی کے قواعد سمجھ سیکے۔		
تشبیہ، استعارہ، تلمیح۔۔۔	1۔ مکمل علم بیان، بدیع اور صنعتوں کی تعریف کر سیکے۔ اشعار میں مثالیں تلاش کر سیکے۔ 2۔ غلط فقرات کی صنائع بدائع کے لحاظ سے درستی کر سیکے۔ بدائع کے لحاظ سے مثالیں تلاش کر سیکے۔ 3۔ کسی فن پارے کا علم بیان کی روشنی میں جائزہ لے سیکے۔ کسی فن پارے میں علم بیان یعنی تشبیہ، استعارہ، کنایہ، مجاز مرسل وغیرہ کی شناخت کر سیکے۔	G12UrU36P110	تشبیہ، تلمیح
مجاز مرسل	1۔ مکمل علم بیان، بدیع اور صنعتوں کی تعریف کر سیکے۔ اشعار میں مثالیں تلاش کر سیکے۔ 2۔ غلط فقرات کی صنائع بدائع کے لحاظ سے درستی کر سیکے۔ 3۔ کسی فن پارے کا علم بیان کی روشنی میں جائزہ لے سیکے۔ کسی فن پارے میں علم بیان یعنی تشبیہ، استعارہ، کنایہ، مجاز مرسل وغیرہ کی شناخت کر سیکے۔	G12UrU37P111	مجاز مرسل حصہ 1
صنائع بدائع	1۔ ادبی تحریروں میں مجازی حوالے سے حسن بیان کی صنعتوں کو ملحوظ رکھ کر انہیں پڑھ سیکے۔ 2۔ مکمل علم بیان، بدیع اور صنعتوں کی تعریف کر سیکے۔ اشعار میں مثالیں تلاش کر سیکے۔ 3۔ غلط فقرات کی صنائع بدائع کے لحاظ سے درستی کر سیکے۔ بدائع کے لحاظ سے مثالیں تلاش کر سیکے۔	G12UrU37P112	مجاز مرسل حصہ 2
		G12UrU38P113	صنائع بدائع حصہ 1
		G12UrU38P114	صنائع بدائع حصہ 2
مرکبات، جملہ سازی	1۔ استحسان کے اصولوں سے آگاہ ہو سیکے۔ عملاً تنقیدی جملے لکھ سیکے۔ 2۔ تنقیدی حسن و قبح کے طریقوں سے آگاہ ہو سیکے۔ شعری و ادبی محضروں/انداز بیان میں تراکیب سازی کو سمجھ سیکے اور مرکبات کو عملاً استعمال کر سیکے۔ 3۔ اصناف سخن کی تعریف کرسکے اور پہچان کر استحسان کر سیکے۔ معنی کے لحاظ سے لغوی اور قرارداد اصطلاحی میں فرق کر سیکے۔	G12UrU39P115	مرکبات، جملہ سازی حصہ 1
		G12UrU39P116	مرکبات، جملہ سازی حصہ 2
روزمرہ، محاورہ، ضرب المثل		G12UrU40P117	روزمرہ، محاورہ، ضرب المثل
اصناف نثر	1۔ اصناف سخن کی تعریف کرسکے اور پہچان کر استحسان کر سیکے۔ معنی کے لحاظ سے لغوی اور قرارداد اصطلاحی میں فرق کر سیکے۔ 2۔ مختصر اصناف کے طرز بیان سے آگاہ ہو سیکے۔ بنیادی اسالیب بیان سے آگاہ ہو سیکے۔	G12UrU41P118	اصناف نثر حصہ 1
		G12UrU41P119	اصناف نثر حصہ 2
اصناف سخن		G12UrU42P120	اصناف سخن حصہ 1
		G12UrU42P121	اصناف سخن حصہ 2
اصناف نظم پارٹ ٹو		G12UrU43P122	اصناف نظم
		G12UrU44P123	روداد حصہ 1
روداد	مہارت: لکھنا 1۔ کھیلوں اور معاشرتی مسائل پر اخبار کے مدیر/متعلقہ محکمہ کو رپورٹ لکھ کر بھیج سیکے۔ 2۔ روزمرہ زندگی کے حوالے سے مفصل روداد، آنکھوں دیکھا حال لکھ سیکے۔ 3۔ روزمرہ زندگی کے تجربات اور مشاہدات کے حوالے سے عمومی مضامین تحریر کر	G12UrU44P124	روداد حصہ 2

	سیکے۔ مہارت: بولنا 1۔ بات کو دہرائے ہوئے اپنے علم اور تجربے کی روشنی میں کمی بیشی کے ساتھ بیان کر سیکے۔ مہارت : انشاء پردازی - کسی بھی موضوع (ادبی، صحافتی، علمی) پر اپنے وسیع مطالعہ کی روشنی میں ترتیب، استدلال، عمدہ مثالوں کے ساتھ منظر نگاری، مکالمہ نگاری، کردار نگاری، کہانی، انشائیہ یا مضمون وغیرہ کی صورت میں تحریر پیش کر سیکے۔ کسی تخلیقی سطح کی تحریر (ڈائری، رپورٹ، تاثر، انشائیہ، کہانی، افسانہ، نظم، غزل وغیرہ) کو مناسب اصلاح کے ساتھ پیش کر سیکے۔		
مکالمہ نویسی	مہارت: سننا سن کر بات/کہانی/مکالمے کی جزئیات کو ترتیب سے یاد رکھ سکے تاکہ بعد ازاں اسے دہرائے کے قابل ہو۔ مہارت: بولنا 1۔ سن کر بات/کہانی/مکالمے کو ترتیب سے دہرا سیکے۔ مہارت : انشاء پردازی 1۔ کسی بھی موضوع (ادبی، صحافتی، علمی) پر اپنے وسیع مطالعہ کی روشنی میں ترتیب، استدلال، عمدہ مثالوں کے ساتھ منظر نگاری، مکالمہ نگاری، کردار نگاری، کہانی، انشائیہ یا مضمون وغیرہ کی صورت میں تحریر پیش کر سیکے۔	G12UrU45P125	مکالمہ نویسی
خط نویسی	مہارت: پڑھنا 1۔ اخبارات اور جرائد میں خبروں، فیچروں، اداروں، رپورٹوں اشتہاروں اور خطوط بنا م مدیر کو ان میں پوشیدہ معنی اور معنی پر اثر انداز ہونے کے حوالے سے سمجھ کر پڑھ سیکے۔ 2۔ دفاتر سے آنے والے تبادلہ و تقرر کے احکام/درخواستیں /مراسلے سمجھ کر پڑھ سیکے۔ مہارت: لکھنا 1۔ اخبارات اور رسائل میں سے کوئی نیوز اسٹوری، خط بنام مدیر یا کوئی مضمون لکھ سیکے۔ نکاح نامہ پر کر سیکے۔ 2۔ کھیلوں اور معاشرتی مسائل پر اخبار کے مدیر/متعلقہ محکمہ کو رپورٹ لکھ کر بھیج سیکے۔ مہارت : مہارت حیات 1۔ روزمرہ زندگی کے مسائل کے حوالے سے غیر رسمی خطوط لکھ سیکے۔ داخلہ فارم، شناختی کارڈ فارم پُر کر سیکے۔ رسید لکھ سیکے۔	G12UrU46P126	خط نویسی
درخواست نویسی	مہارت: پڑھنا 1۔ دفاتر سے آنے والے تبادلہ و تقرر کے احکام/درخواستیں /مراسلے سمجھ کر پڑھ سیکے۔ مہارت: لکھنا 1۔ تبادلہ و ترقی کے لیے محکمانہ درخواست لکھ سیکے۔	G12UrU47P127	درخواست نویسی
آپ بیتی۔۔۔	1۔ روزمرہ زندگی کے حوالے سے مفصل روداد، آنکھوں دیکھا حال، آپ بیتی یا مفصل مضمون کی صورت میں لکھ سیکے۔ 2۔ کم از کم پندرہ ہزار الفاظ و تراکیب کا احاطہ کر سیکے۔	G12UrU48P128	آپ بیتی۔۔۔
رموزاوقاف		G12UrU49P129	رموزاوقاف حصہ 1

	شاعری میں متعلقہ اصطلاحات کے تحت درج نکات کی شناخت کر سکے۔ شاعری میں مطلع، مقطع، ردیف، قافیہ، وزن، سادہ بحر کی شناخت کر سکے۔	G12UrU49P130	رموز اوقاف حصہ 2
مطلع، مقطع، ردیف، قافیہ		G12UrU50P131	قواعد تعارف، ردیف، قافیہ
		G12UrU50P132	مطلع، مقطع
مضمون نویسی	1۔ کسی مضمون کے اجزا اور خصوصیات، اصطلاحات اور تصورات کے حوالے سے سمجھ کر پڑھ سکے۔ 2۔ کوئی مضمون لکھتے ہوئے علمی حاصلات اور نکات پر اپنی رائے کے ساتھ پیش کر سکے۔ 3۔ اخبارات اور رسائل میں سے کوئی نیوز اسٹوری، خط بنام مدیر یا کوئی مضمون لکھ سکے۔ 4۔ حالات حاضرہ معاشی و سیاسی مسائل پر اپنا نقطہ نظر تحریر کر سکے۔ 5۔ روزمرہ زندگی کے حوالے سے مفصل روداد، آنکھوں دیکھا حال، آپ بیتی یا مفصل مضمون کی صورت میں لکھ سکے۔ 6۔ روزمرہ زندگی کے تجربات اور مشاہدات کے حوالے سے عمومی مضامین تحریر کر سکے۔ 7۔ کسی بھی موضوع (ادبی، صحافتی، علمی) پر اپنے وسیع مطالعہ کی روشنی میں ترتیب، استدلال، عمدہ مثالوں کے ساتھ منظر نگاری، مکالمہ نگاری، کردار نگاری، کہانی، انشائیہ یا مضمون وغیرہ کی صورت میں تحریر پیش کر سکے۔	G12UrU51P133	مضمون نویسی

g. Pakistan Studies

Unit	SLO	Package Code	Package Name
Genesis Republic of Pakistan	Ideology of Pakistan and It's Concepts	G12PSU01P01	Ideology of Pakistan
	Components	G12PSU01P02	Fundamentals of Pakistan Ideology
	Islam		
	Social Justice, Equality and Responsibility		
	Basic Human Rights		
	Pakistan- The Result of Muslim Struggle	G12PSU01P03	The Outcome of Sustained Muslim Effort
	Foundation of Muslim League,		
	Ali-Garh Tehreek		
	Tehreek-e-Khilafat	G12PSU01P04	Khilafat Movement
	Pakistan Resolution	G12PSU01P05	Pakistan Resolution
	Elections of 1945-46	G12PSU01P06	Elections 1945 and 1946
	Cabinet Mission Plan		
	Interim Government	G12PSU01P07	Interim Government
	3rd June 1947 Plan,		
	Cripp's Mission	G12PSU01P08	Cripps Mission, Simla Conference and Indian Independence Act
	Simla Conference		
	Independence Act 1967		
	Emergence of Pakistan		
Initial problems of Islamic republic of Pakistan	A . Problems	G12PSU02P09	Early Problems
	Administrative Problems		
	Introductory Remarks		
	Red Cliff Awards-Its Injustices	G12PSU02P10	Injustice by Radcliffe and Accession of Disputes of States
	Influx of Refugees/ Division of Assets		
	Division of Armed Forces and Assets		
	Water/ Canal Dispute		
	Accession of Princely States of Junagadh		
	Manavadar,Kashmir,Hyderabad Deccan		
	Actions Taken to Resolve the Problem	G12PSU02P11	National Solidarity
	National Solidarity and Stability		
	Physiography of Pakistan	G12PSU02P12	Physiography of Pakistan
	Climatic Zones of Pakistan		

	Imbalanced Economic Growth and Regional Disparities	G12PSU02P13	Imbalanced Economic Growth and Regional Disparities
	Influence of Climate on Human Life	G12PSU02P14	Geo-Strategic Position, Climate Effect, and Transit Facilities
	Geo-Strategic Position		
	Gate Way to Land Located Countries : Afghanistan and Central Asian Republic		
	influence of climate on human life	G12PSU02P15	Tourist Attraction
	attraction for tourists		
	geo-strategic position	G12PSU02P16	Map Reading
	map reading		
	Significance and impact objectives resolution 1949	G12PSU02P17	Objective Resolution and Islamic Provision
	Islamic provisions in the constitutional documents		
	step islamization since 1949	G12PSU02P18	Islamization, Rights and Responsibilities in Pakistan
	rights and responsibilities		
	Affects : if one branches and deviates from them	G12PSU02P19	Human Rights and Last Sermon of Hazrat Muhammad (P.B.U.H)
	human rights		
	salient features of world human rights declaration 1948		
	khutba hujjat ul vida the comprehensive charter of human rights		
	Federal Government - various Institutions and their function	G12PSU02P20	Composition of the Federal Legislative
	Provincial Governments and their functions.		
	Local Governments with reference to Devolution plan 2001		
	Islamic perspective of good governance	G12PSU02P21	Concept of Good Governance in Islam
	Administration in the era of Hazrat Umar (R.A.U)..	G12PSU02P22	System of Government of Hazrat Omar (R.A) Part 1
		G12PSU02P23	System of Government of Hazrat Omar (R.A) Part 2
	Devolution of power Plan and Good Governance.	G12PSU02P24	Devolution of Power Plan and Obstacles in the Way
	Obstacles and remedies in the way to Good		

culture and islamic republic of Pakistan	Meaning and significance of culture	G12PSU03P25	Definition and Importance of Culture
	Cultural Heritage of Pakistan: main sites.	G12PSU03P26	Important Ancient Sites of Cultural Significance
	Common Characteristics of Pakistan Culture.	G12PSU03P27	Gender Equality
	Gender equity; issues and problems.		
	Rights of women in Islam.	G12PSU03P28	Rights of Women
	Ancient Civilization with reference to Pakistan epitome from history of world by Tryon be	G12PSU03P29	Ancient Civilization of Pak, Be Pakistani and Buy Pakistani
	Be Pakistani, buy Pakistani		
	Policies for women participation in socio-political affairs of Pakistan		
languages of Islamic republic of Pakistan	Languages as vehicle of cultural integration and human relationship	G12PSU04P30	Regional Languages in Pakistan: Urdu and Balochi
	the National Language Urdu historical development		
	Regional languages in Pakistan: Balochi		
	Regional languages in Pakistan: Punjabi	G12PSU04P31	Regional Languages in Pakistan: Punjabi, Pushto and Sindhi
	Regional languages in Pakistan: Pushto		
	Regional languages in Pakistan: Sindhi		
national integration and prosperity	Meaning of national integration and cohesion.	G12PSU05P32	Importance of National Integration
	Significance in an Islamic Democratic State		
	Problems in the way to national cohesion and international in Pakistan	G12PSU05P33	Problems of National Integration
	Remedial measures. My, your and our role.		
	Remedial measures. My, your and our role.	G12PSU05P34	Measures to Overcome Problems in National Integration
economic planning and development in Islamic republic of Pakistan	meaning and importance of economic planning and development	G12PSU06P35	Economic Planning and Agricultural Development
	agriculture and development		
	Industrial Development.	G12PSU06P36	
	Trade and Commerce.		

	Natural Resource Development and their conservation		Industrial Development, Natural Resources, Trade and Commerce
	Industrial Development.	G12PSU06P37	Industrial Development, I.T, Trade and Commerce
	Trade and Commerce.		
	Information Technology		
	Economic Planning in the light of all five years plans:	G12PSU06P38	Economic Planning in Light of Five Years Plan
	Education and Health for all	G12PSU06P39	Economic Planning, Health and Education for All
	Strategy		
	achievement		
Foreign policy of Islamic republic of Pakistan	Definition of Foreign policy	G12PSU07P40	Basic Principle of Foreign Policy
	Principles of the Foreign policy of Pakistan		
	Principles of the Foreign policy of Pakistan	G12PSU07P41	Aims and Objectives of Foreign Policy
	Objectives of the Foreign Policy of Pakistan		
	Determinants of the foreign policy of Pakistan	G12PSU07P42	Foreign Policy of Pakistan and Iran
	Foreign Policy of Foreign Policy of Iran,		
	India.	G12PSU07P43	Foreign Policy of Pakistan and India
	USA,	G12PSU07P44	Foreign Policy of Pakistan, USA and Afghanistan
	Afghanistan,		
	Saudi Arabia.	G12PSU07P45	Foreign Policy of Pak and SA: Its Failures and Successes
	A glimpse of World		
	Affairs and our Foreign		
	Policy successes and failures		

5. How to Use This Document in Class

This guide is designed to make lesson planning and classroom delivery easier. Teachers can use it in the following ways:

- **Lesson Planning:** Before each lesson, check the tables to see which video(s) cover the relevant SLO(s) for your class.
- **Classroom Instruction:** Play the appropriate video during class to illustrate concepts, reinforce learning, or introduce new topics.
- **Curriculum Alignment:** Ensure videos are paired with textbook units to maintain alignment with the Pakistan National Curriculum.

- **Revision and Reinforcement:** Use videos to review previous lessons or provide extra practice for students who need it.
- **Multi-Grade or Mixed-Level Classes:** Teachers can identify videos suitable for different grade levels and adjust classroom activities accordingly.
- **Assessment:** Each video package has a corresponding assessment, named with the same package code, which can be found in the accompanying Excel sheets. Teachers can use these assessments to check students' understanding immediately after showing the video or during lesson follow-ups.

The document is structured to make it easy to locate videos quickly, so you spend more time teaching and less time searching.

6. Ethical and Copyright Considerations

All digital content provided through this platform is procured by the Ministry of Federal Education and Professional Training and aligns with the Pakistan National Curriculum. When using these resources, teachers should keep the following points in mind:

- **Authorized Use Only:** Videos are for educational purposes within classrooms or official teaching activities. Do not distribute, copy, or upload content to public platforms without permission.
- **Attribution:** When using content in presentations or shared materials, credit the Ministry as the source.
- **Respect for Learners:** Ensure all content is used in a way that supports inclusive and respectful learning environments.
- **No Modification Without Approval:** Do not alter video content, subtitles, or audio tracks unless officially authorized.
- **Data Privacy:** Avoid sharing student information or recordings when using videos in digital or remote learning settings.

Following these guidelines ensures that teachers use the digital content responsibly while maintaining legal and ethical standards.

7. Conclusion

This guide is designed to make using digital learning videos simple, efficient, and aligned with the Pakistan National Curriculum and Student Learning Outcomes. By following the mapping tables and video codes, teachers can quickly identify the right resources for each lesson, support student learning effectively, and plan classroom activities with confidence.

We encourage teachers to explore the videos, pair them with assessments, and integrate them into lesson plans to create engaging and interactive learning experiences.

For any questions, technical issues, or additional support, please contact the Federal Directorate of Education (FDE)'s content team or the designated focal points.

Empower your students, explore the digital resources, and make every lesson an engaging step toward learning success.